

The Erdős–Straus workbench

a reader of its results, with proofs

An audit and reading prepared by Claude (Anthropic),
model `claude-opus-5-5` (Opus 5.5), at maximum reasoning effort

Workbench: The Clankers (KokunoYumeto and the contributors credited below)

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Abstract

The Erdős–Straus workbench of The Clankers (KokunoYumeto/erdos-strauss-foundation and the Zenodo collection 10.5281/zenodo.20401937) studies the equation $4/n = 1/x + 1/y + 1/z$. Its main documents run to more than a thousand pages:

- a 619-page cumulative archive (title page dated 30 August 2026) and an earlier 488-page reader (7 August);
- a 67-page and an 86-page supplement on bounded congruence transport (8 and 9 September);
- a 124-page reader of 20 September;
- a continuation of 17–23 September with 28 structural items.

This reader says what these documents establish, where each result is proved, and how far each was checked here.

Its core (Sections 3–4) is proved in full here:

- the divisor criterion for a fixed first denominator, which is the archive’s divisor-shell bijection, with the halved range of the least denominator added;
- the first two residual shells, and the fact that a prime whose first two shells are both empty lies in one of the six classes modulo 840 that the classical reduction leaves open;
- the Jacobi-symbol barrier, with a sharper group-theoretic form and a partial converse;
- the least-class function, whose strict records are recomputed; an independent computation finds no further strict record below $3 \cdot 10^9$;
- a general fixed-divisor construction, of which the archive’s four explicit families are instances.

Section 2 maps the rest: every section of the 619-page archive with its page range and its bearing on the equation, and the supplements and the continuation. It also determines the exact scope of one theorem of the supplement, on shear transport between adjacent shells. The theorem holds for every prime $p \not\equiv 2 \pmod{3}$, and no weaker congruence hypothesis suffices. Among the primes $p \equiv 2 \pmod{3}$ it fails only on a minority, of upper relative density at most 0.335 (5.24% below 10^7).

One consequence concerns the archive’s largest computation, its carrier census. Every carrier the census uses is a polynomial identity on a residue class. By the obstruction of Schinzel and Yamamoto, no such identity covers a class that is a square. So at least $1/1024$ of the census’s final domain can never be forced by carriers of this kind.

Section 6 states the negative results with their exact scope. The workbench does not claim a proof of the conjecture, and nothing here proves or disproves it.

About this reader

The workbench, and who made it. The collective author name of the workbench is *The Clankers*. The workbench credits:

- `u/CommonCareful3149`, for the mod-107 arithmetic-progression observation;
- `u/UmbrellaCorp_HR`, for arithmetic, positivity-code and prime-production contributions and for the central-angle and inversive-circle theorems of the archive’s §6;

- u/CivQ17, for the diagrams behind result R05;
- u/Far-Lifeguard519, for the initiating computational investigation;
- u/TextBackground496 and u/JGPTech, for source ideas;
- the owner, KokunoYumeto (u/lepthymo), for the research directions and examples, for directing and developing the investigations with AI, and for the proof and source-fidelity requirements.

These credits are from README.md, “Attribution and collaboration”, and CONTRIBUTIONS.md. The archive records language and computational models as contributing-model provenance for particular calculations, not as authors (archive, “Research status and provenance”).

What was read.

- The repository at commit d20b32e (23 September 2026), the latest on 26 September.
- The 488-page reader and its source, which are in the repository.
- The 619-page archive (00_ERDOS_STRAUSS_Project_Reader.pdf) and its source archive (01_..._Source_Verification_and_Machine_Memory.zip), downloaded from the Zenodo record 10.5281/zenodo.22678971.

Locators have the form “archive §*n*, p. *m*” for the 619-page PDF, “supplement §*n*” for the 86-page supplement, and file names and line numbers for the repository. The audit notes cited as 21_, 43_, 43a_, 47_ and 47a_ are on the branch `claude/claude-ab-grind-20260925` of KokunoYumeto/zeta-function-research-reader, in `contrib/claude-ab/grind_20260925/`.

How it was checked.

- The core of Sections 3–4 was re-derived by the author of this reader. Most of it was refereed three times as part of an earlier reader of four workbenches [11] (Appendix B).
- The new computations of this reader were made with its own scripts: the record primes of the least-class function, the deficit sets of the record shells, the four families, and the exact scope of one theorem of the supplement (Appendix B).
- This version includes the findings of one independent referee pass, by a Claude instance that wrote none of the text. It read the reader for overstatement, for understatement and missed generality, and for missed or implied results. Each finding was verified before it was applied, by reading the archive at the cited place or by an independent script. One of the referee’s numbers was corrected in the process (Appendix B).
- Everything else carries a status label saying how far it was checked.

Status labels.

- Proved here: the proof is given in full in this reader.
- Generalised here: the workbench’s statement is proved here under weaker hypotheses than the workbench states.
- Audited: the workbench’s proof was read in full, every step was checked, and independent checks were run.
- Audited (certificate): the finite certificate on which a proof rests was recomputed independently, and the deduction from it was read.
- Audited in part: the parts named were checked; the rest is located.

- **Reproduced:** the finite content of a statement (its examples, tables, identities or counts on a stated range) was recomputed by an independent script; its general proof was not re-derived.
- **Located:** the statement and its proof were found and are described; nothing was checked.

The workbench itself labels statements as proved, certificate-checked, conjectural or withdrawn. Its label is quoted where it differs from the one above.

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1 The problem, what is known, and what the workbench claims

1.1 The equation

The Erdős–Straus conjecture asserts that for every integer $n \geq 2$ there are positive integers x, y, z with

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Repeated denominators are allowed; for example $4/5 = 1/2 + 1/4 + 1/20$. It suffices to treat primes: $4/2 = 1/1 + 1/2 + 1/2$, and if $4/p = 1/x + 1/y + 1/z$ then $4/(mp) = 1/(mx) + 1/(my) + 1/(mz)$.

1.2 What is known

The following results frame everything below. Each is cited from the literature; none is re-proved here, except where Section 3 proves a special case.

- *The classes modulo 840.* Explicit polynomial identities solve the equation for every n in a primitive class $r \bmod 840$ unless r is a square modulo 840. So for primes only the classes

$$p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$$

remain. Elsholtz and Tao attribute this to Obláth and to Mordell [2, 6]; Proposition 3.5 below proves the reduction for primes.

- *Why identities cannot finish the job.* Elsholtz and Tao [2] call a primitive class $n \equiv r \pmod{q}$ *solvable by polynomials* if there are polynomials P_1, P_2, P_3 which take positive integer values for all sufficiently large n in the class and satisfy $4/n = 1/P_1(n) + 1/P_2(n) + 1/P_3(n)$. A primitive class with r a square modulo q is not solvable by polynomials. They cite Mordell and Schinzel for this, and trace the reciprocity argument behind it to Schinzel and to Yamamoto ([2], §1).
 - Section 3.4 gives the workbench’s shell form of this obstruction, with proof.
 - Proposition 4.6 gives a form that applies to the carriers of the archive’s census.
- *Density.* The number of $n < N$ for which the equation has no solution is at most $N \exp(-c \log^{2/3} N)$ for an absolute $c > 0$ and N large (Vaughan [9], as stated by Elsholtz and Tao [2]).
- *Counting.* Elsholtz and Tao bound the number $f(n)$ of solutions and its averages, and classify solutions at primes as Type I (p divides exactly one denominator) or Type II (p divides two) [2].
- *Verification.* The conjecture has been checked for all $n \leq 10^{14}$ (Swett, cited in [2]) and for all $n \leq 10^{17}$ (Salez [8]); a 2025 preprint reports the bound 10^{18} [4].

1.3 What the workbench claims

The repository’s README opens by saying that the project claims neither a proof nor a counterexample. The research state calls it an ongoing programme, not a completed proof. The archive’s status section (p. 4) names the remaining global obligation: for every relevant prime, at least one marked shell class must be shown to be occupied. The workbench’s scope statements are careful and, in the parts read here, accurate. Where a workbench statement was found to be wrong or overstated, this reader says so and gives the correction (Section 6 and Appendix A).

1.4 The approach in one paragraph

Fix a prime p and a candidate least denominator a with $p/4 < a < p$. The remaining equation $1/y + 1/z = 4/p - 1/a$ has solutions exactly when a^2 has a divisor in one of two prescribed residue

classes modulo $R = 4a - p$ (Theorem 3.1). The workbench calls a a *shell* and R its *residual*. For $p \equiv 1 \pmod{4}$ the residual is $\equiv 3 \pmod{4}$, and the shells in play are $R = 3, 7, 11, \dots$ up to about p . Most of the workbench’s arithmetic is organised around three kinds of tool:

- sufficient conditions under which a shell is occupied, stated as congruence conditions on p or on the prime factors of a ;
- obstructions under which a shell is empty;
- bounded “carrier” censuses, which count how much of a finite space of residue data these conditions cover.

Around this arithmetic core the documents also develop geometric and operator-theoretic constructions (inversive circles, Lorentz lifts, Boolean carriers, an auxiliary torus, a Busy-Beaver map). Section 2 places each of them and says how it bears on the equation.

2 The documents, and what each contains

2.1 The editions and how they relate

Edition	Where	Contents and relation to the others
Project reader, 488 pages, 7 August 2026	repository root, with source and Lean archives	The earlier edition. The README says it is preserved and is not the newest reader.
Cumulative archive, 619 pages; title page dated 30 August 2026	Zenodo 22666493 and 22678971, file 00	The 488-page reader continued. Its main source has the same section skeleton, now 60,561 lines instead of 57,452. Fourteen satellite files add §§10–19, the $Q = 83$ step of §24 (Theorem 24.241) and the carrier refinements §§24.1–24.3. It begins from the source of the earlier Zenodo record 21845035, version 69 (archive p. 3). It is the front file of the Zenodo collection.
Bounded congruence transport, 67 pages, 8 September	Zenodo 22666493, files 02–03	The 58-statement supplement.
Expanded supplement, 86 pages, 9 September	Zenodo 22678971, files 07–08; repository research/expanded-2026-09-08/	The same, with 14 further statements: 72 statements in 15 source files, with checkers (§2.3).
ES/Fable reader, 124 pages, 20 September	repository research/incoming/es-fable-zeta-bridge-20260920/	A coordinate bridge from Erdős–Straus solutions to the Fable cover and a weighted conductor of the zeta programme (§2.4).
Continuation, 17–23 September	repository README, “Start reading” (28 items); research/incoming/	41 dated packages; the results bench R01–R10 and the integration arguments X1–X7 (§2.5).

These editions have different contents; the repository preserves them rather than overwriting them (README, edition table).

2.2 The 619-page archive, section by section

Page numbers are those of the PDF. The column “Bearing” says whether a section bears on the equation:

- *direct*: it proves or tests solvability;

- *literature*: it audits a published approach to the equation;
- *side*: a construction not about solvability.

The status column refers to this reader.

§	pp.	Content	Bearing	Status here
front	3–5	Research status: the census figures of §24, the four families, the record-shell results, and an explicit list of non-claims	summary	read
1	6–16	Scope, road map, mathematical status, attribution, lineage	summary	read in part
2	16–18	The discriminant-five identity $(2n+1)^2 - 5 = 4(n^2 + n - 1)$ and its root involution, from u/CivQ17	side	located
3	18–20	The shell $p_* = 8,803,369$, $R_* = 107$, reached through Busy-Beaver values ($S(4) = 107$, $\text{Space}(5) = 12289$)	direct (one shell)	$R = 107$ is the least occupied shell at p_* (Proposition 4.3)
4	20–22	Alpöge’s constant-Jacobian map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with a three-point fibre	side	see §7
5	22	Withdrawn constructions (a cube-root interpolation on the Fable fibre), with the point of failure	side	read
6	22–27	Central-angle and inversive-circle theorems of u/UmbrellaCorp_HR; a Lorentz reflection	side	located
7	27–31	The complete fibre of F over $(-\frac{1}{4}, 0, 0)$ and a marked-factor lift	side	see §7
8	31–38	The singular C_1 – C_2 – C_3 support object; dihedral actions	side	located
9	38–93	The exact divisor-shell bijection (Theorem 9.1, for every odd prime) and its layer count (Theorem 9.64, p. 90); then directed blade transport and a quarter-radius inversion	direct (Theorems 9.1 and 9.64), side (the rest)	the bijection and the count re-proved here (Theorem 3.1(b))
10	93–98	Residual sieve modulo 840: an occupied shell with $R = 3$ or $R = 7$ at every prime $p \equiv 1 \pmod{24}$ outside the six hard classes (Theorem 10.6); the Jacobi-symbol barrier (Theorem 10.9)	direct	proved here (Propositions 3.4, 3.6)
11	98–100	Elsholtz–Tao Type I and Type II coordinates mapped to the residual shell	literature	located
12	100–106	Adjacent programmes (López, Dyachenko, Bradford): what each proves, and where a claimed bridge fails	literature	located
13	106–114	Mordell’s constructions and Rosati’s theorem; Salez’s seven charts; Bright–Loughran surfaces; the Vaughan boundary	literature	located
14	114–119	López’s structural solutions: exact divisor coordinates and a quantifier repair	literature	located
15	119–124	The Star–Kneser programme: a one-way forcing theorem and a retracted converse	direct	located
16	124–127	The mod-107 deficit progression, following u/CommonCareful3149	direct	progression and its corollary reproduced (§4.1)

§	pp.	Content	Bearing	Status here
17	127–135	Two-fraction criteria; Monks–Velingker; Ionascu–Wilson classes, with the printed record list recovered and one printed class label corrected (Proposition 17.6); Zelator’s prime-square classification	literature	records recomputed (§4.1)
18	135–148	The residual-eleven support ideal, its colons, and the paired $R = 7/R = 11$ affine theorem	direct	located
19	148–157	Strict record shells: an open conjecture and the boundary example $p = 806,521$	direct	records and deficits reproduced (§4.1)
20	157–179	The shell $p = 12289$, $R = 11$, $a = 3075$: counts $(3, 6, 3)$, its dihedral orbit, and the Star–Kneser quotient certificate	direct (one shell)	counts reproduced (below the table); a is the least occupied shell; the rest located see §7
21	179–200	The split-zero semiring $G(R) = R \sqcup \{\tau\}$ and the C_1 carrier	side	
22	200–201	Rigidified denominator towers over $\mathbb{F}_1$	side	located
23	201–250	Mirror completion of the Boolean carrier; the 12289 orbit matrix	side	located
24 (i)	250–298	The opening of §24: Tomita–Takesaki transport and C_3 modular data (Theorems 24.1–24.44)	side	located
24 (ii)	299–604	The rest of §24, from Theorem 24.45: shell arithmetic. • Star–Kneser forcing (Theorems 24.45–24.48, pp. 299–301); • endpoint divisor fibres (Theorems 24.50–24.63, pp. 303–312); • coupled and adjacent shells, and the base of the census (Theorem 24.100, p. 335); • the carriers for $R = 11$ to $R = 31$, including the constant-factor carriers (Lemma 24.218, p. 422) and the four families (Theorem 24.219, p. 423); • the $R = 71$, $R = 191$ and $Q = 83$ steps (Theorems 24.233, 24.237 and 24.241, pp. 436–444); • §§24.1–24.3 (pp. 451–477): the $R = 167$ and $R = 239$ carrier theorems and the refresh of the $R = 167$ coefficient; • §24.4 (pp. 478–604): a return to the $R = 23$ lineage, interleaved with further side constructions.	direct (with side material in §24.4)	four families proved here (Theorem 4.5); census located (§4.3)
25	605–616	A map from Busy-Beaver extrema to finite shell carriers	side	see §7

At $p = 12289$ the shell $a = 3075$ has

$$E_a = \{41, 1845, 15375\}, \quad M_a = \{5, 225, 1875, 5043, 42025, 1891125\}.$$

So $(|E_a|, |M_a|, |E_a|) = (3, 6, 3)$, and exactly six unordered solutions have least denominator 3075 (recomputed here).

So about 250 of the 619 pages bear directly on the equation:

- §§3, 9 (Theorems 9.1 and 9.64), 10, 15, 16 and 18–20, about 70 pages;
- §24 from p. 299 to the end of §24.3 (p. 477), 179 pages.

Parts of the 127 pages of §24.4 also do. The literature audits of §§11–14 and 17 add about 35 pages. The rest is side constructions. The archive itself describes their connection to the equation as an interface still to be supplied by an explicit typed map (archive p. 4).

2.3 The bounded-transport supplement (72 statements)

The supplement starts from the primes $p = 12h + 1$ and all shells $3h + 1 \leq a \leq 9h$ of Theorem 3.1. Its sections are:

1. original shells, coordinates and integer reconstruction, with the first-two-shell sieve and an all-shell no-hit sieve;
2. local congruences with all integral lifts retained, with an integral cyclic bridge;
3. bounded divisibility and simultaneous congruences: unary Boolean transport, a shared-variable Chinese-remainder construction credited to a colleague, raw-scale hit incidence, and a boundary-swap completion;
4. exact cross-shell transport at a fixed rational ratio;
5. a primitive Euclidean successor with its full arithmetic domain;
6. arithmetic localisation along the original norm subdivision, with a reading of Connes’ primitive intersections;
7. a strictly decreasing denominator with exact arithmetic return;
8. the Navier–Stokes auxiliary cover: the torus matrix $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ of the announced Navier–Stokes manuscript, its covering fibres and character corrections, and sampling and reconstruction estimates.

The supplement’s own README states that its results prove or disprove neither the Erdős–Straus conjecture nor the Riemann hypothesis nor the Yang–Mills mass gap, and that they do not validate the Navier–Stokes construction.

Status. Reproduced in part. A second-pass audit, interrupted by a usage limit before it wrote its report, recomputed the finite content of the statements of Sections 1–7. Its scripts, which are independent of the workbench’s code, found every checked statement correct:

- the gluing and Chinese-remainder theorems on random systems and on all shells of small primes;
- the ratio-transport domain on all 57,882 ordered shell pairs of the primes $p \equiv 1 \pmod{12}$ below 260;
- the shear and no-two-shears theorems on all shells of 131 primes below 4000;
- the Connes marked successor on all primes $p \equiv 1 \pmod{12}$ below $3 \cdot 10^5$;
- the fixed- y successor on all 3,114 source witnesses below 1500;
- the orchard partition on all 42 admissible triples below 1300;
- the boundary four-candidate law on all 72,282 raw records below 400;
- the unary box, the input/output incidence and the raw-scale theorems on their stated ranges.

One of its checks, on the smallest member of a boundary orbit, reported mismatches. The script expected the pattern $(\min(A, B), \max(A, B))$ in every case. The supplement’s own table (boundary_swap_completion.tex, lines 296–309) gives a one-point orbit’s own point as its smallest member, which is what the script found. The mismatch is therefore in the script, not in the supplement.

The same pass also looked at scope, and its search led to Proposition 2.1 below. The general proofs were not re-derived, except that of Section 5’s no-two-shears theorem; the auxiliary-torus Section 8 was not checked. The scripts are in `checks/es_reader/second_pass_partial/`.

Section 5 of the supplement studies the two unimodular shears $L(m, n) = (m + n, n)$ and $J(m, n) = (m, m + n)$ as maps between adjacent shells. It encodes the solutions with first denominator a by the set

$$\mathcal{W}_a = \{(m, n) \in \mathbb{Z}_{>0}^2 : m \mid pa, n \mid pa, \gcd(m, n) = 1, R_a \mid m + n\},$$

which is in bijection with the ordered solutions (a, y, z) through $y = pak/n$, $z = pak/m$, $k = (m + n)/R_a$ (supplement §1, first lemma; this is Theorem 3.1(b) in other coordinates). A *shear edge* is a step $(a, m, n) \mapsto (a + 1, L(m, n))$ or $(a, m, n) \mapsto (a + 1, J(m, n))$ with $(m, n) \in \mathcal{W}_a$ and the image in $\mathcal{W}_{a+1}$. The supplement proves, for $p = 12h + 1$ and shells $3h + 1 \leq a \leq 9h$, that no two shear edges compose (Theorem “No two consecutive positive Euclidean shell steps”, `publication.tex`, line 933). Its proof uses the hypothesis on p at one step only, and there only through $p \equiv 1 \pmod{3}$.

Proposition 2.1 (the no-two-shears theorem, exact scope). *Let p be a prime, and consider all shells a with $p/4 < a < p$.*

- (a) *If $p \not\equiv 2 \pmod{3}$, no two shear edges compose: there is no path $(a, m, n) \mapsto (a + 1, m', n') \mapsto (a + 2, m'', n'')$ of shear edges.*
- (b) *Such paths exist at some primes $p \equiv 2 \pmod{3}$. At $p = 131$ one is $(33, 1, 33) \mapsto (34, 1, 34) \mapsto (35, 1, 35)$, with $R_a = 1, 5, 9$. At $p = 1613$, where $p \equiv 1 \pmod{4}$, one is $(404, 1, 404) \mapsto (405, 1, 405) \mapsto (406, 1, 406)$, with $R_a = 3, 7, 11$. Proposition 2.2 says exactly at which primes.*

Status. Generalised here, with the workbench’s proof. The supplement states (a) for $p \equiv 1 \pmod{12}$ and $3h + 1 \leq a \leq 9h$ only. Checked on the full graph of every prime below 2000 (script `esr_twoshears.py`: no path for $p \equiv 1 \pmod{3}$; 14 paths for $p \equiv 2 \pmod{3}$), and by a complete search up to $p \leq 10^7$ that uses the shape of an edge proved below: it finds 19,558 paths of two L -edges, all at primes $p \equiv 2 \pmod{3}$, and as many of two J -edges.

Proof. The shape of one edge (the supplement’s classification, whose proof needs only $a + 1 < p$). Let $(a, m, n) \mapsto (a + 1, m + n, n)$ be an L -edge. Then n divides pa and $p(a + 1)$, whose greatest common divisor is p . If $n = p$, then $p \nmid m$ by coprimality, and $m + p$ divides $p(a + 1)$ and is prime to p , so $m + p \mid a + 1$; but $m + p > p > a + 1$. So $n = 1$. If $p \mid m$, then $m + 1$ is prime to p and exceeds $a + 1$, and yet divides $p(a + 1)$, which is impossible; so $p \nmid m$ and $m \mid a$. Then $m + 1 \leq a + 1 < p$, so $m + 1 \mid a + 1$. Finally $R_a \mid m + 1$ and $R_{a+1} = R_a + 4 \mid m + 2$. For a J -edge, exchange m and n ; the exchange preserves every $\mathcal{W}_a$.

(a) An L -edge ends at $(a + 1, m + 1, 1)$, whose first coordinate is at least 2; a J -edge starts at a pair whose first coordinate is 1. So an L -edge is never followed by a J -edge, and by the exchange a J -edge is never followed by an L -edge. Suppose two L -edges pass through $(a + i, m + i, 1)$ for $i = 0, 1, 2$. By the shape of each edge, $m + i \mid a + i$ and $R_a + 4i \mid m + i + 1$ for $i = 0, 1, 2$. Put $K = a - m$. Then $m + i \mid K$ for each i , and one of $m, m + 1, m + 2$ is divisible by 3, so $3 \mid K$. Also $R_a + 4i$ divides $4(m + i + 1) - (R_a + 4i) = 4m + 4 - R_a =: C$, and one of $R_a, R_a + 4, R_a + 8$ is divisible by 3, so $3 \mid C$. But $C = 4m + 4 - R_a + p = p + 4 - 4K \equiv p + 1 \pmod{3}$. Hence $p \equiv 2 \pmod{3}$. Two J -edges are excluded by the exchange.

(b) At $p = 131$: $R_{33} = 1$ divides $1 + 33$; $R_{34} = 5$ divides 35 ; $R_{35} = 9$ divides 36 ; each coordinate divides pa , and each pair is coprime. At $p = 1613$ (prime): $3 \mid 405$, $7 \mid 406 = 7 \cdot 58$ and $11 \mid 407 = 11 \cdot 37$. Each step is $J(1, n) = (1, n + 1)$. $\square$

So the hypothesis $p \equiv 1 \pmod{12}$ of the supplement can be weakened to $p \not\equiv 2 \pmod{3}$. The next proposition shows that no weaker congruence hypothesis suffices, although the conclusion

fails only at a minority of the primes $p \equiv 2 \pmod{3}$ (Remark 2.3). The primes $p \equiv 2 \pmod{3}$ are solvable in any case (Section 3); both propositions concern the transport graph, not solvability.

Proposition 2.2 (the shear graph). *Let p be a prime. Consider the directed graph whose vertices are the triples (a, m, n) with $p/4 < a < p$ and $(m, n) \in \mathcal{W}_a$, and whose arrows are the shear edges.*

- (a) *Every vertex has at most one outgoing and at most one incoming edge. So every connected component is a directed path.*
- (b) *Diagonal paths. Let $R = 4a - p$, and let $k \geq 1$ with $a + k < p$. The vertices $(a + i, a + i, 1)$, $0 \leq i \leq k$, form a path of L -edges if and only if $R + 4i$ divides $p + 4$ for every $0 \leq i \leq k$.*
- (c) *If $(a + i, m + i, 1)$, $0 \leq i \leq k$, is a path of L -edges and $R = 4a - p$, then $R + 4i$ divides $p + 4$ for every $0 \leq i \leq k - 1$. In particular a path of two edges forces $R(R + 4) \mid p + 4$.*
- (d) *Every prime $p \equiv 131 \pmod{180}$ has a path of two edges. More generally, let $q \geq 1$ and let c be prime to q , with $c \equiv 2 \pmod{3}$ if $3 \mid q$ (these are exactly the classes that contain primes $\equiv 2 \pmod{3}$). Then infinitely many primes $p \equiv c \pmod{q}$ have a path of two edges. So if Proposition 2.1(a) holds at all but finitely many primes of a union of classes modulo q , then none of those classes contains primes $\equiv 2 \pmod{3}$.*
- (e) *For every $k \geq 1$, infinitely many primes have a path of length k .*

Status. Proved here. Found by the referee pass of this reader; each part checked here. Part (a) was confirmed on the full graph of every prime below 1200. A complete enumeration below 10^7 through (c) and the shape of an edge finds:

- 19,558 two-edge paths at 17,425 primes;
- $R(R + 4) \mid p + 4$ on every one of those paths;
- all 13,822 primes $p \equiv 131 \pmod{180}$ among the primes with a path.

The least primes with a path of length 1, 2, 3, 4 and 5 are 11, 131, 3461, 109,391 and 1,183,451. The referee gave 3,533,141 for length 5; it has such a path, but it is not the least.

Proof. (a) By the shape of an edge, an L -edge starts at a vertex with $n = 1$ and a J -edge at one with $m = 1$. So the only possible successors of (a, m, n) are $(a + 1, m + 1, 1)$, when $n = 1$, and $(a + 1, 1, n + 1)$, when $m = 1$. Both are possible only if $m = n = 1$. Then $(1, 1) \in \mathcal{W}_a$ forces $R_a \mid 2$, so $R_a = 1$. Either successor would then need $R_{a+1} = 5$ to divide 3, which is false. The only possible predecessors of $(a + 1, m', n')$ are $(a, m' - 1, 1)$, by an L -edge when $n' = 1$, and $(a, 1, n' - 1)$, by a J -edge when $m' = 1$. Both are possible only if $m' = n' = 1$, and then $m' - 1 = 0$ is not a positive coordinate. Finally, a increases along edges, so there are no cycles.

(b) The triple $(a + i, a + i, 1)$ satisfies the divisibility and coprimality conditions of $\mathcal{W}_{a+i}$. It lies in $\mathcal{W}_{a+i}$ exactly when $R + 4i \mid a + i + 1$. As $R + 4i$ is odd and $4(a + i + 1) = (R + 4i) + (p + 4)$, this holds exactly when $R + 4i \mid p + 4$. Consecutive vertices are joined by an L -edge, since $L(a + i, 1) = (a + i + 1, 1)$.

(c) Let $0 \leq i \leq k - 1$ and put $K = a - m$. The edge from the i -th vertex gives $m + i + 1 \mid a + i + 1$, hence $m + i + 1 \mid K$. The i -th vertex lies in $\mathcal{W}_{a+i}$, so $R + 4i \mid m + i + 1$. Hence $R + 4i \mid K$. Also $R + 4i$ divides $4(m + i + 1) - (R + 4i) = 4m + 4 - R$. Adding $4K$, it divides $4a + 4 - R = p + 4$. For two edges, R and $R + 4$ are odd with difference 4, hence coprime, so $R(R + 4) \mid p + 4$.

(d) If $p \equiv 131 \pmod{180}$, then $p \equiv 3 \pmod{4}$, so $R = 1$ occurs, at $a = (p + 1)/4$. Also $45 \mid p + 4$, so 1, 5 and 9 divide $p + 4$, and (b) applies with $k = 2$. In general, choose an odd R with the following properties:

- $R \equiv -c \pmod{4}$ if $4 \mid q$;
- no prime $\ell \geq 5$ dividing q divides $R(R + 4)(R + 8)$;

- the one of $R, R+4, R+8$ that is divisible by 3 is not divisible by 9.

Such an R exists by the Chinese remainder theorem: each prime $\ell \geq 5$ excludes three classes, and modulo 9 the classes 0, 1 and 5 are excluded. Put $m' = R(R+4)(R+8)$. Its three factors are odd with pairwise differences 4 and 8, hence pairwise coprime, and $\gcd(m', q) \mid 3$. The congruences $p \equiv c \pmod{q}$, $p \equiv -4 \pmod{m'}$ and $p \equiv -R \pmod{4}$ are compatible: modulo 3 both of the first two give $p \equiv 2$, and modulo 4 the first and the third agree by the choice of R . Together they define a class prime to its modulus, which contains infinitely many primes by Dirichlet's theorem. Each such prime $p > R+8$ has the diagonal path of (b) with $a = (p+R)/4$ and $k = 2$. If $3 \nmid q$, every class prime to q contains primes $\equiv 2 \pmod{3}$ (by the Chinese remainder theorem and Dirichlet's theorem). If $3 \mid q$, a class contains such primes exactly when $c \equiv 2 \pmod{3}$.

(e) Take $R = 1$. The modulus $\text{lcm}(1, 5, 9, \dots, 4k+1)$ is odd, so the congruences $p \equiv 3 \pmod{4}$ and $p \equiv -4$ modulo it define a class prime to its modulus. Each prime $p > (4k+1)/3$ in it has, by (b), a diagonal path of length k from $a = (p+1)/4$. $\square$

Remark 2.3 (how often the theorem fails). Among the primes $p \equiv 2 \pmod{3}$, those with a path of two edges have upper relative density at most 0.335. So the conclusion of Proposition 2.1(a) holds on a set of primes $p \equiv 2 \pmod{3}$ of lower relative density at least 0.665. Below 10^7 it fails at 17,425 of the 332,384 primes $p \equiv 2 \pmod{3}$ (5.24%); the smallest primes $p \equiv 2 \pmod{3}$ at which it holds are 2, 5, 11, 17, 23, 29, 41, 47, 53, 59, 71, 83, ...

Proof. By Proposition 2.1(a) and Proposition 2.2(c), a prime with a path of two edges satisfies:

- $p \equiv 2 \pmod{3}$;
- for some odd R with $R(R+4) \leq p+4$, both $p \equiv -4 \pmod{R(R+4)}$ and $p \equiv -R \pmod{4}$.

Fix R and put $m = R(R+4)$. By Dirichlet's theorem, the second condition picks out a set of relative density $\delta_R = (1 + [3 \mid m]) / (2\varphi(m))$ among the primes $p \equiv 2 \pmod{3}$. For large R the bound must be uniform in x :

- For $4m \leq x^{1/2}$. The Brun–Titchmarsh inequality $\pi(x; Q, b) \leq 2x / (\varphi(Q) \log(x/Q))$ for $Q < x$ [5], with $Q = 4m$, bounds the number of such primes up to x by $2x / (\varphi(m) \log x)$. This is $(4 + o(1)) / \varphi(m)$ times the number of primes $p \equiv 2 \pmod{3}$ up to x .
- For $4m > x^{1/2}$. Each class has at most $x/m + 1$ members up to x , and $R \leq \sqrt{x+4}$. The total over these R is $O(x^{3/4})$.

Since $\sum_R 1/\varphi(R(R+4))$ converges, letting $x \rightarrow \infty$ and then the cut between small and large R tend to infinity bounds the upper relative density by $\sum_{R \text{ odd}} \delta_R$. Numerically, the partial sum over $R < 2 \cdot 10^5$ is 0.334275. The rest is less than $2 \cdot 10^{-4}$, by $n/\varphi(n) < e^\gamma \log \log n + 3/\log \log n$ for $n \geq 3$ [7]. So the sum is less than 0.335. $\square$

2.4 The 124-page ES/Fable reader

It encodes a solution of $4/p = 1/x + 1/y + 1/z$ as the binary quartic

$$H_{\text{ES}}(U, V) = -S^{-1}(U - pV)(U - xV)(U - yV)(U - zV), \quad S = p + x + y + z,$$

with normalised coefficients $u_0 = -1/S$, $u_1 = 1$, $u_2 = -(p(x+y+z) + xy + yz + zx)/S$, $u_3 = (p(xy + yz + zx) + xyz)/S$ (which equals $5xyz/S$ on solutions) and $u_4 = -pxyz/S$. It then transports the quartic to the Fable cover and to a weighted conductor of the zeta programme.

Proposition 2.4. (a) On solutions, $p = -5u_4/u_3$ and

$$625u_0u_4^3 - 125u_3u_4^2 + 25u_2u_3^2u_4 - 4u_3^4 = 0.$$

(b) Conversely, let u_0, u_2, u_3, u_4 be complex numbers with $u_0 u_3 u_4 \neq 0$ that satisfy this relation. Put

$$S = -1/u_0, \quad p = -5u_4/u_3, \quad P = u_3 S/5, \quad \sigma_1 = S - p, \quad \sigma_2 = -u_2 S - p\sigma_1.$$

Then the roots x, y, z of $T^3 - \sigma_1 T^2 + \sigma_2 T - P$ are non-zero and satisfy $4/p = 1/x + 1/y + 1/z$. The quartic $-S^{-1}(U - pV)(U - xV)(U - yV)(U - zV)$ has the coefficients $u_0, 1, u_2, u_3, u_4$.

Status. Proved here. The README of the package states (a), and 21_ (Proposition 21.6) also checked the relation. Part (b) was found by the referee pass of this reader and is proved here.

Proof. (a) Write $\sigma_1 = x + y + z$, $\sigma_2 = xy + yz + zx$ and $P = xyz$. The equation is $4P = p\sigma_2$, so the third elementary symmetric function of p, x, y, z is $p\sigma_2 + P = 5P$; this gives $u_3 = 5P/S$, and $p = -5u_4/u_3$ follows. Substituting and multiplying by $S^4/625$, the left side becomes $P^3(p^3 - p^2S + p(p\sigma_1 + \sigma_2) - 4P) = P^3(p\sigma_2 - 4P) = 0$.

(b) With the stated definitions, $u_0 = -1/S$, $u_4 = -pu_3/5$, $u_2 = -(p\sigma_1 + \sigma_2)/S$ and $u_3 = 5P/S$. Substituting, and using $S = p + \sigma_1$, gives

$$625u_0u_4^3 - 125u_3u_4^2 + 25u_2u_3^2u_4 - 4u_3^4 = \frac{u_3^3}{S}(5p^3 - 5p^2S + 5p(p\sigma_1 + \sigma_2) - 20P) = \frac{5u_3^3}{S}(p\sigma_2 - 4P).$$

So the relation gives $p\sigma_2 = 4P$. Now $xyz = P \neq 0$, since $u_3 \neq 0$, and $p \neq 0$, since $u_4 \neq 0$. Hence $1/x + 1/y + 1/z = \sigma_2/P = 4/p$. For the quartic, the coefficients of U^4 , U^3V , U^2V^2 , UV^3 and V^4 in $-S^{-1}(U - pV)(U - xV)(U - yV)(U - zV)$ are:

- $-1/S = u_0$;
- $(p + \sigma_1)/S = 1$;
- $-(p\sigma_1 + \sigma_2)/S = u_2$;
- $(p\sigma_2 + P)/S = 5P/S = u_3$;
- $-pP/S = u_4$.

□

So the relation is the equation itself, written in these coordinates: by (a) and (b), it cuts out exactly the quartics that come from triples, possibly complex, with $4/p = 1/x + 1/y + 1/z$. The Fable and conductor maps built on it are located; their comparison with the zeta programme is in the zeta audit (47_ overlap OV-13).

2.5 The continuation of 17–23 September

The README's "Start reading" list has 28 items with proof packages in research/incoming/. The results bench R01–R10 and the integration arguments X1–X7 accompany them.

- The first-pass audit 43a_ lists them all.
- Section 5 states the eleven items audited there.
- Section 6 gives their negative results.

3 The core arithmetic, with proofs

Throughout, p is an odd prime and a *solution* is a triple of positive integers with $4/p = 1/x + 1/y + 1/z$. The workbench mostly works with $p = 12h + 1$ ($h \geq 1$). The results of this section, except Proposition 3.6, were first stated in the reader of four workbenches [11], whose Erdős–Straus section was refereed three times (Appendix B); the audit note is 43_ and the first-pass report 43a_.

3.1 The shell criterion

For an integer $a > p/4$ not divisible by p put $R_a = 4a - p$, $S_a = pa$ and

$$E_a = \{u : u \mid a^2, R_a \mid 4u + 1\}, \quad M_a = \{u : u \mid a^2, R_a \mid u + a\}.$$

A shell a is occupied if $E_a \cup M_a \neq \emptyset$.

Theorem 3.1 (shell criterion). (a) If $x \leq y \leq z$ is a solution, then $p/4 < x \leq 3p/4$. More precisely, $x < p/2$ and $x < y$, unless $p \equiv 3 \pmod{4}$ and $(x, y, z) = (\frac{p+1}{2}, \frac{p+1}{2}, \frac{p(p+1)}{4})$. For $p = 12h + 1$ this gives $3h + 1 \leq x \leq 6h$.

(b) For each integer $a > p/4$ not divisible by p (in particular each a with $p/4 < a < p$), the ordered pairs (y, z) of positive integers with $1/a + 1/y + 1/z = 4/p$ number exactly $2|E_a| + |M_a|$. A pair with $y = z$ occurs if and only if $R_a = 1$ (possible only for $p \equiv 3 \pmod{4}$ and $a = (p + 1)/4$); otherwise $|M_a|$ is even and the unordered pairs number $|E_a| + \frac{1}{2}|M_a|$. For $p \equiv 1 \pmod{4}$ this is always the case.

(c) Hence the equation is solvable at p if and only if some shell a with $p/4 < a < p/2$ is occupied.

Status. Proved here. Where each part is in the workbench:

- The bijection of (b). Archive Theorem 9.1 (“Exact divisor-shell bijection”, p. 38) states it for every odd prime p and every $R \equiv -p \pmod{4}$ with $\gcd(R, pa) = 1$. That is exactly the generality of (b), since $\gcd(R, pa) = 1$ holds if and only if $p \nmid a$.
- The count $2|E_a| + |M_a|$. Archive Theorem 9.64 (p. 90) splits the divisors by their p -adic layer and proves by an involution that the two outer layers have equal size. The targets $-4a^2$, $-a$ and -4^{-1} are the archive’s table (394) (§10, pp. 93–94, stated there for $R \equiv 3 \pmod{4}$).
- The version for $p = 12h + 1$. The supplement’s ES-01/ES-02 (§1), audited in the first pass, state the criterion for $p = 12h + 1$ and $a \in [3h + 1, 9h]$.

This reader adds (a), with its exceptional solution at $p \equiv 3 \pmod{4}$; the analysis of the swap in (b), that is, when $y = z$ occurs, the parity of $|M_a|$ and the unordered count; and (c). The halved range at $p \equiv 1 \pmod{12}$, with x unique, is already a reduction step of the workbench’s item 22, checked by reading in the first pass (43a_§2.22), and items 6 and 7 work in it; the workbench’s later packages attribute these coordinates to Elsholtz and Tao [2], §2. Checked against a brute-force count of the ordered pairs on all 4,726 pairs (p, a) with $p < 260$ an odd prime and $p/4 < a < p$ (a pair with $y = z$ exactly in the 29 cases with $R_a = 1$) and on all 4,303 pairs with $p < 120$, $p/4 < a \leq 3p$ and $p \nmid a$; on all 4,398 shells of the primes $p \equiv 1 \pmod{12}$ below 700, and in the first pass on all 8,124 shells below 1000; (a) against all 2,259 solutions at the odd primes below 400; every prime $p \equiv 1 \pmod{12}$ below 30,000 has an occupied shell.

Proof. (a) The other two terms are positive, so $1/x < 4/p$; and x is the least denominator, so $4/p \leq 3/x$. Hence $p/4 < x \leq 3p/4$. Suppose $x > p/2$, so $x \geq (p + 1)/2$ as p is odd, and $x < p$. Then $2/y \geq 1/y + 1/z = 4/p - 1/x > 2/p$, so $y < p$. From $4xyz = p(xy + yz + zx)$ the prime p divides xyz , and as $0 < x \leq y < p$ it divides $z = pz'$; then $xy(4z' - 1) = pz'(x + y)$ with $p \nmid xy$, so $4z' - 1 = pk$ with $k \geq 1$. Put $N = pk + 1 = 4z'$; then $1/x + 1/y = 4k/N$, that is, $(4kx - N)(4ky - N) = N^2$ with both factors positive. But $4kx - N \geq 2k(p + 1) - N = N + 2(k - 1)$, and likewise for y ; so $k = 1$ and $x = y = N/2 = (p + 1)/2$, with $z = p(p + 1)/4$, which needs $p \equiv 3 \pmod{4}$. Finally, if $x < p/2$ and $y = x$, then $1/x + 1/y > 4/p$, which is impossible. For $p = 12h + 1$ the integers in $(p/4, p/2)$ are $3h + 1, \dots, 6h$.

(b) Write $R = R_a$, $S = S_a$. Then $R \geq 1$ is odd, because p is. Since $p \nmid a$, $\gcd(a, R) = \gcd(a, p) = 1$ and $\gcd(p, R) = \gcd(p, 4a) = 1$, so S is a unit modulo R . The equation $1/y + 1/z = R/S$ is equivalent to $(Ry - S)(Rz - S) = S^2$, and both factors are positive because $1/y$ and $1/z$ are smaller than R/S . So the ordered pairs correspond bijectively to ordered factorisations $S^2 = dd'$ into positive divisors with $d \equiv d' \equiv -S \pmod{R}$, through $y = (d + S)/R$, $z = (d' + S)/R$. If $d \mid S^2$ and $d \equiv -S$, then d is a unit and $d' = S^2/d \equiv S^2(-S)^{-1} = -S$; so the pairs are counted by $\#\{d \mid S^2 : d \equiv -S$

(mod R). Since $\gcd(p, a) = 1$, each divisor of $S^2 = p^2 a^2$ is uniquely $d = p^i v$ with $i \in \{0, 1, 2\}$ and $v \mid a^2$. Using $4a \equiv p \pmod{R}$: for $i = 2$, $p^2 v \equiv -pa \iff pv \equiv -a \iff 4pv \equiv -p \iff R \mid 4v + 1$, so these v form E_a ; for $i = 1$, $p v \equiv -pa \iff R \mid v + a$, so these v form M_a ; and $d \mapsto S^2/d$ maps the layer $i = 2$ bijectively onto the layer $i = 0$ while preserving the congruence. The total is $2|E_a| + |M_a|$. The swap $(y, z) \mapsto (z, y)$ is $d \mapsto S^2/d$; it maps each layer i to the layer $2 - i$, so a fixed point lies in the layer $i = 1$, where it is $v \mapsto a^2/v$ with only possible fixed point $v = a$. That needs $R \mid 2a$, i.e. $R \mid 2$ (as $\gcd(R, a) = 1$), i.e. $R = 1$. If $R \neq 1$ the swap has no fixed point, $|M_a|$ is even, $y \neq z$, and the unordered pairs number $|E_a| + \frac{1}{2}|M_a|$. For $p \equiv 1 \pmod{4}$, $R \equiv -p \equiv 3 \pmod{4}$, so $R \neq 1$.

(c) For $p \equiv 1 \pmod{4}$ it follows from (a) and (b). For $p \equiv 3 \pmod{4}$ both sides hold: the shell $a = (p + 1)/4$ lies in $(p/4, p/2)$ and has $R_a = 1$, so $u = a$ lies in M_a . $\square$

The workbench also gives the exact bookkeeping around this criterion:

- an explicit bijection between a tagged set and the ordered solutions, with its inverse read off from the p -adic valuation of $Ry - S$ (archive (526), p. 151; supplement §1);
- the global statement that a counterexample prime would satisfy the finite, computable condition $E_a = M_a = \emptyset$ for every a .

Both were re-derived in the first pass (43a_ §1.2).

3.2 The first two shells

Proposition 3.2 (the residual-3 shell). *Let $a = 3h + 1 = (p + 3)/4$, so $R_a = 3$. Let $P_1 = \prod (2v_\ell(a) + 1)$ over the primes $\ell \equiv 1 \pmod{3}$ dividing a , and P_2 the same product over the primes $\ell \equiv 2 \pmod{3}$. Then $E_a = M_a = \{u \mid a^2 : u \equiv 2 \pmod{3}\}$ and $|E_a| = |M_a| = P_1(P_2 - 1)/2$. So a solution with least denominator a exists if and only if a has a prime factor $\equiv 2 \pmod{3}$, and there are then exactly $3P_1(P_2 - 1)/4$ unordered solutions with least denominator a .*

Status. Proved here; the workbench’s residual-3 sieve (supplement §1, `first_two_shell_sieve.tex`; archive Theorem 10.3, “Exact $R = 3$ factor criterion”, p. 94, for $p \equiv 1 \pmod{24}$), audited. Checked on all 4,472 primes $p \equiv 1 \pmod{12}$ below $2 \cdot 10^5$, and in the first pass on all 19,564 below 10^6 .

Proof. Since $a \equiv 1 \pmod{3}$, $3 \nmid a$. Modulo 3 both $4u + 1$ and $u + a$ are $\equiv u + 1$, so both conditions say $u \equiv 2 \pmod{3}$. A divisor $u = \prod \ell^{e_\ell}$ of a^2 has $u \equiv (-1)^{\sum e_\ell} \pmod{3}$, the sum over the primes $\ell \equiv 2 \pmod{3}$. Their exponent vectors $(0 \leq e_\ell \leq 2v_\ell(a))$ number P_2 , and the difference between the numbers with even and with odd sum is $\prod_\ell \sum_{e=0}^{2v_\ell} (-1)^e = 1$; so $(P_2 - 1)/2$ have odd sum, and the other primes contribute the factor P_1 . By Theorem 3.1(b) the unordered solutions containing a number $\frac{3}{2}P_1(P_2 - 1)/2$, and $a = 3h + 1$ is the least denominator of each of them because it is the smallest possible one. Finally $P_2 > 1$ exactly when a has a prime factor $\equiv 2 \pmod{3}$. (The count is an integer: $a \equiv 1 \pmod{3}$ makes the total exponent of those primes even, so $P_2 \equiv 1 \pmod{4}$.) $\square$

Proposition 3.3 (the residual-7 shell). *Let $a = 3h + 2 = (p + 7)/4$, so $R_a = 7$ and $7 \nmid a$. Let n_3 be the total exponent in a of the primes $\equiv 3 \pmod{7}$ and n_{24} that of the primes $\equiv 2$ or $4 \pmod{7}$. The shell is occupied if and only if a has a prime factor $\equiv 5$ or $6 \pmod{7}$, or $n_3 \geq 3$, or $n_3 \geq 1$ and $n_{24} \geq 1$.*

Status. Proved here; the workbench’s residual-7 sieve, audited, together with its count as coefficients of $\prod_\ell \sum_{e=0}^{2v_\ell(a)} T^{\lambda(\ell)e}$ in $\mathbb{Z}[T]/(T^6 - 1)$. Checked on all 4,472 primes $p \equiv 1 \pmod{12}$ below $2 \cdot 10^5$ (and below 10^6 in the first pass).

Proof. The exterior condition $7 \mid 4u + 1$ is $u \equiv -4^{-1} \equiv 5 \pmod{7}$, and the middle condition is $u/a \equiv -1 \equiv 3^3 \pmod{7}$; in base 3 the discrete logarithms of $1, \dots, 6$ are $0, 2, 1, 4, 5, 3$. *Sufficiency.* A prime $\ell \equiv 5$ dividing a gives $u = \ell$; a prime $\ell \equiv 6$ gives $u = a\ell$. If $n_3 \geq 3$, a product of primes

$\equiv 3$ dividing a^2 with total exponent 5 gives $u \equiv 3^5$. If primes $t \equiv 3$ and $\ell \in \{2, 4\} \pmod{7}$ divide a , then $u = alt$ (when $\ell \equiv 2 = 3^2$) or $u = a\ell/t$ (when $\ell \equiv 4 = 3^4$) divides a^2 and has $u/a \equiv 3^3$. *Necessity.* If none of the conditions holds, every prime factor of a is $\equiv 1, 2, 3$ or 4 , and either $n_3 = 0$ or $(1 \leq n_3 \leq 2$ and $n_{24} = 0)$. If $n_3 = 0$, a and all divisors of a^2 lie in the subgroup $\{1, 2, 4\}$ of squares, which contains neither 5 nor -1 . Otherwise $a \equiv 3^{n_3}$ and a divisor of a^2 is $\equiv 3^e$ with $0 \leq e \leq 2n_3 \leq 4$; the exterior target needs $e \equiv 5 \pmod{6}$ and the middle one $e - n_3 \equiv 3 \pmod{6}$ with $|e - n_3| \leq 2$, both impossible. $\square$

The shells with $R = 3$ and $R = 7$ exist exactly for the primes $p \equiv 1 \pmod{4}$, with $a = (p+3)/4$ and $a = (p+7)/4$ (Generalised here; checked on all 1,611 such primes below $3 \cdot 10^4$). Proposition 3.3 holds for all of them, since its proof never uses $p \equiv 1 \pmod{3}$. Proposition 3.2 holds as stated for $p \equiv 1 \pmod{12}$ only. For $p \equiv 5 \pmod{12}$, $a = (p+3)/4 \equiv 2 \pmod{3}$, so modulo 3 the middle condition reads $u \equiv 1$; then $E_a = \{u \mid a^2 : u \equiv 2\}$ and $M_a = \{u \mid a^2 : u \equiv 1\}$, with $|E_a| = P_1(P_2 - 1)/2$ and $|M_a| = P_1(P_2 + 1)/2$ by the parity count of the proof above, and there are exactly $|E_a| + \frac{1}{2}|M_a| = P_1(3P_2 - 1)/4$ unordered solutions with least denominator a (an integer: the total exponent in a of the primes $\equiv 2 \pmod{3}$ is odd, so $P_2 \equiv 3 \pmod{4}$). That shell is always occupied ($1 \in M_a$), and its occupancy criterion holds trivially, since $a \equiv 2 \pmod{3}$ has a prime factor $\equiv 2 \pmod{3}$; those primes are solvable anyway, since $p \equiv 2 \pmod{3}$.

Proposition 3.4 (the two-shell survivors). *Let $p \equiv 1 \pmod{12}$ be prime. Both shells $a = (p+3)/4$ and $a = (p+7)/4$ are unoccupied if and only if the following three conditions hold:*

- $p \equiv 1 \pmod{24}$;
- every prime factor of $(p+3)/4$ is $\equiv 1 \pmod{3}$;
- every prime factor of $(p+7)/4$ is $\equiv 1, 2$ or $4 \pmod{7}$.

Every such p is a square modulo 7 and satisfies $p \equiv \pm 1 \pmod{5}$. So it lies in one of the six hard classes $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$.

Status. Proved here. The conclusion modulo 840 is archive Corollary 10.7 (p. 95), the contrapositive of its Theorem 10.6; the first pass stated the congruence mod 24 as an observation on its computation. Checked on all primes $p \equiv 1 \pmod{12}$ below 10^6 . Of these, exactly 989 have both shells unoccupied. All 989 lie in the six hard classes, and each class occurs; the least in the classes 1, 121, 169, 289, 361, 529 are 2521, 12721, 2689, 1129, 1201 and 3049.

Proof. If $p \equiv 13 \pmod{24}$, then $(p+3)/4$ is even and $2 \equiv 2 \pmod{3}$, so the first shell is occupied (Proposition 3.2). If $p \equiv 1 \pmod{24}$, the first shell is unoccupied exactly when every prime factor of $(p+3)/4$ is $\equiv 1 \pmod{3}$. Also $(p+7)/4$ is even, so $n_{24} \geq 1$, and Proposition 3.3 says that the second shell is occupied exactly when $(p+7)/4$ has a prime factor $\equiv 3, 5$ or $6 \pmod{7}$, a non-residue. If it has none, $(p+7)/4$ is a square modulo 7, and so is $p \equiv 4 \cdot \frac{p+7}{4}$. For the congruence modulo 5:

- if $p \equiv 2 \pmod{5}$, then $5 \mid (p+3)/4$ and $5 \equiv 2 \pmod{3}$, so the first shell is occupied;
- if $p \equiv 3 \pmod{5}$, then $5 \mid (p+7)/4$ and 5 is a non-residue modulo 7, so the second shell is occupied;
- $p = 5$ is not $\equiv 1 \pmod{24}$.

So $p \equiv \pm 1 \pmod{5}$, and $p \equiv 1, 2, 4 \pmod{7}$, and $p \equiv 1 \pmod{24}$. These are the unit squares modulo 840. $\square$

3.3 Which primes remain

The workbench proves solvability for $p \equiv 2 \pmod{3}$, $p \equiv 3 \pmod{4}$ and $p \equiv 13 \pmod{24}$ (scope identities that it does not advertise as new) by the families $(p, \frac{p+1}{3}, \frac{p(p+1)}{3})$, $(\frac{p+1}{4}, \frac{p(p+1)}{2}, \frac{p(p+1)}{2})$ and $(a, \frac{pa}{2}, pa)$ with $a = \frac{p+3}{4}$ (even when $p \equiv 13 \pmod{24}$); so its own reduction leaves the primes $p \equiv 1 \pmod{24}$ (archive Proposition 10.2). Its later packages restrict to the *hard* primes $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$, which are exactly the unit squares modulo 840 (a group of order $1 \cdot 1 \cdot 2 \cdot 3 = 6$). The reduction to them is classical (Obláth and Mordell; see Elsholtz–Tao [2] and Mordell [6]). The 22 September packages use it without proof.

The archive proves a stronger form (Theorem 10.6, “Non-square modulo 840 theorem”, p. 95). For every prime $p \equiv 1 \pmod{24}$ outside the six hard classes, the shell with $R = 3$ or the shell with $R = 7$ is occupied. Its proof uses explicit divisors in these two shells; Proposition 3.4 above is the same statement. The classical proof uses shells up to $R = 15$ and five identities. It is given here for comparison.

Proposition 3.5 (the reduction modulo 840). *Let $p \equiv 1 \pmod{24}$ be prime, and let (A, B, R) be $(1, 2, 7)$, $(2, 1, 7)$, $(1, 1, 7)$, $(1, 2, 15)$ or $(2, 1, 15)$ according as $p \equiv 3, 5, 6 \pmod{7}$ or $p \equiv 2, 3 \pmod{5}$. Then $D = (p + R)/(4AB)$ and $C = (A + pB)/R$ are positive integers and*

$$\frac{4}{p} = \frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD}.$$

So the equation is solvable at every prime $p \equiv 1 \pmod{24}$ that is not a square modulo 840.

Status. Proved here (the reduction is classical); the first pass covered the 18 non-square unit classes $\equiv 1 \pmod{24}$ by the same kind of families. The identities hold for all integers in the stated classes, not only for primes. Checked on all primes $p \equiv 1 \pmod{24}$ below $3 \cdot 10^5$.

Proof. With $RC = A + pB$ and $4ABD = p + R$: $\frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD} = \frac{pC + pB + A}{pABCD} = \frac{C(p+R)}{pABCD} = \frac{p+R}{pABD} = \frac{4}{p}$. Since $p \equiv 1 \pmod{8}$, 8 divides $p + 7$ and $p + 15$, so $4AB \mid p + R$; $R \mid A + pB$ is the stated congruence modulo 7 or 5, together with $A + pB \equiv A + B \equiv 0 \pmod{3}$ when $R = 15$. The primes $p \equiv 1 \pmod{24}$ left over have $p \equiv 1, 2, 4 \pmod{7}$ and $p \equiv \pm 1 \pmod{5}$, which are exactly the unit squares modulo 840. $\square$

3.4 The Jacobi-symbol barrier

Proposition 3.6 (the Jacobi-symbol barrier). *Let p be an odd prime, $R \equiv 3 \pmod{4}$, $a = (p + R)/4$ an integer, and $\gcd(R, pa) = 1$. If the Jacobi symbol (q/R) equals 1 for every prime $q \mid a$, then the shell a is unoccupied. The converse is false: at $p = 5$, $R = 7$, $a = 3$ the only prime factor of a is a non-residue modulo 7, and yet the shell is unoccupied.*

Status. Proved here; archive Theorem 10.9 (“Jacobi-symbol barrier”, p. 96) and Proposition 10.10 (“The Jacobi barrier has no converse”, pp. 96–97), whose proofs are correct as read.

Proof. The hypothesis $\gcd(R, pa) = 1$ gives $p \nmid a$: otherwise p would divide $4a - p = R$. So, by the proof of Theorem 3.1(b), the shell is occupied exactly when some divisor d of $S^2 = p^2a^2$ satisfies $d \equiv -pa \pmod{R}$. Let $\chi = (\cdot/R)$ be the Jacobi symbol, a character on the units modulo R . The hypothesis gives $\chi(a) = 1$, and $p \equiv 4a \pmod{R}$ gives $\chi(p) = \chi(4)\chi(a) = 1$. So $\chi(d) = 1$ for every divisor d of p^2a^2 , while $\chi(-pa) = \chi(-1) = (-1)^{(R-1)/2} = -1$ because $R \equiv 3 \pmod{4}$. So no divisor lies in the target class. For the converse: the squares modulo 7 are 1, 2, 4, so $(3/7) = -1$; the divisors of $15^2 = 225$ are $\equiv 1, 2, 3, 4, 5 \pmod{7}$, and the target $-15 \equiv 6$ is not among them. $\square$

So at a prime $p \equiv 1 \pmod{4}$, a shell with residual R can be occupied only if $a = (p + R)/4$ has a prime factor that is a non-residue modulo R in the Jacobi sense. The argument is a quadratic-character argument: every divisor inherits the residue condition. This is the kind of mechanism behind the obstruction to polynomial identities recalled in Section 1. The barrier explains why the workbench’s occupancy criteria, such as Propositions 3.2 and 3.3, are phrased through non-residue prime factors of a .

The quadratic character can be replaced by the whole group that the divisors generate. This gives a sharper barrier, and a converse when the exponents of a are large enough. Both were found by the referee pass of this reader; the proofs below were checked here.

Proposition 3.7 (the group barrier). *Let p be an odd prime, $R \geq 3$ with $R \equiv -p \pmod{4}$, $a = (p+R)/4$ an integer, and $\gcd(R, pa) = 1$. Let $G_a \leq (\mathbb{Z}/R)^\times$ be the subgroup generated by 4 and the primes dividing a .*

- (a) *If $-1 \notin G_a$, the shell a is unoccupied.*
- (b) *This contains Proposition 3.6. For prime $R \equiv 3 \pmod{4}$ the two hypotheses are equivalent. For composite R the barrier (a) is strictly stronger: at $p = 53$, $a = 26$, $R = 51$ one has $G_a = \langle 2 \rangle = \{1, 2, 4, 8, 13, 16, 26, 32\}$, which does not contain -1 , while $(2/51) = -1$.*

Proof. (a) Every divisor u of a^2 , and a itself, lie in G_a . The two targets -4^{-1} and $-a$ lie in G_a exactly when -1 does, because $4, a \in G_a$. So if $-1 \notin G_a$, no divisor hits a target. (b) If $(q/R) = 1$ for every prime $q \mid a$, then G_a lies in the kernel of the Jacobi character, which does not contain -1 when $R \equiv 3 \pmod{4}$. For prime $R \equiv 3 \pmod{4}$ the group $(\mathbb{Z}/R)^\times$ is cyclic of order $2 \cdot \frac{R-1}{2}$ with $\frac{R-1}{2}$ odd. Its element -1 is the only element of order 2. So $-1 \notin G_a$ exactly when G_a has odd order, that is, when G_a lies in the subgroup of squares, which has odd order $\frac{R-1}{2}$. That is the hypothesis of Proposition 3.6. At $R = 51$: 2 has order 2 modulo 3 and 8 modulo 17, hence order 8 modulo 51, with the powers listed. Also $13 = 2^6$ and $4 = 2^2$ modulo 51, so $G_{26} = \langle 2 \rangle$. Finally $(2/51) = (2/3)(2/17) = -1$. $\square$

Proposition 3.8 (a converse under an exponent condition). *In the setting of Proposition 3.7, let H_a be the subgroup generated by the primes dividing a . Suppose that $2v_\ell(a) \geq \text{ord}_R(\ell) - 1$ for every prime $\ell \mid a$.*

- (a) *The residues modulo R of the divisors of a^2 are exactly the elements of H_a .*
- (b) *The shell a is occupied if and only if $-1 \in H_a$ or $-4^{-1} \in H_a$.*
- (c) *If moreover R is a prime $\equiv 3 \pmod{4}$, the shell is occupied if and only if some prime factor of a is a non-residue modulo R .*

Proof. (a) A divisor of a^2 is $\prod \ell^{e_\ell}$ with $0 \leq e_\ell \leq 2v_\ell(a)$. Under the hypothesis, e_ℓ runs through a complete set of residues modulo $\text{ord}_R(\ell)$, so ℓ^{e_ℓ} runs through all of $\langle \ell \rangle$. The products then fill the subgroup generated by these cyclic subgroups, which is H_a . (b) The targets are -4^{-1} and $-a$, and $-a \in H_a$ exactly when $-1 \in H_a$, since $a \in H_a$. (c) If every prime factor of a is a residue, Proposition 3.6 applies. Suppose some prime factor is a non-residue. The group $(\mathbb{Z}/R)^\times$ is cyclic of order $R-1 = 2 \cdot \frac{R-1}{2}$ with $\frac{R-1}{2}$ odd. So every subgroup of odd order lies in its unique subgroup of order $\frac{R-1}{2}$, which is the subgroup of squares. Since H_a contains a non-square, its order is even. It therefore contains the unique element of order 2, which is -1 , and the shell is occupied by (b). $\square$

Status. Both proved here. Checked on all 123,538 shells with $R \geq 3$ of the primes $5 \leq p < 1500$: the group barrier holds on every one of them. On 9,462 of them, all with composite $R \equiv 3 \pmod{4}$, it applies where the Jacobi barrier does not. The criterion of Proposition 3.8(b) is correct on all 677 shells that satisfy the exponent condition. For prime $R \equiv 3 \pmod{4}$ the two barriers agree on all 18,157 shells of the primes

$p \equiv 1 \pmod{4}$ below 1500. The counterexample $p = 5, R = 7, a = 3$ to the converse of Proposition 3.6 fails the exponent condition: $\text{ord}_7(3) = 6 > 2v_3(3) + 1 = 3$.

4 Least classes, record primes, explicit families, and the carrier census

4.1 The least-class function and its record primes

The archive (§17.4 and §19, pp. 131 and 148–157) puts, for $i \geq 1$, following Ionascu and Wilson [3],

$$C_i = \left\{ n \geq 2 : \frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \text{ with } x \leq \frac{n + 4i - 1}{4} \right\}, \quad h(n) = \min\{i : n \in C_i\}.$$

An odd prime p is a *strict record* if $h(q) < h(p)$ for every prime $q < p$.

Lemma 4.1. *Let $p \equiv 1 \pmod{4}$. The least denominator occurring in any solution at p is the least occupied shell a . With $R = 4a - p$ one has $h(p) = (R + 1)/4$. For $p \equiv 3 \pmod{4}$, $h(p) = 1$, and $h(2) = 1$.*

Status. Proved here.

Proof. If a is occupied, some solution has a as a denominator, so the least denominator of that solution is at most a . Conversely, the least denominator x of a solution is itself an occupied shell: it exceeds $p/4$ (Theorem 3.1(a)) and is completed by the other two denominators. So the least occupied shell equals the least denominator over all solutions. A solution lies in C_i exactly when $4x - p \leq 4i - 1$. For $p \equiv 1 \pmod{4}$ the residual $4x - p$ is $\equiv 3 \pmod{4}$, so the least i is $(R + 1)/4$. For $p \equiv 3 \pmod{4}$ the shell $(p + 1)/4$ is occupied (Theorem 3.1(c)). For $p = 2$, take $4/2 = 1/1 + 1/2 + 1/2$ with $x = 1 \leq 5/4$. $\square$

Corollary 4.2. *If a prime p does not lie in one of the six hard classes 1, 121, 169, 289, 361, 529 modulo 840, then $h(p) \leq 2$. Hence every strict record after 73 lies in a hard class.*

Status. Proved here; found by the referee pass of this reader. It is the archive’s Theorem 10.6 and Proposition 10.2 read through Lemma 4.1. An independent computation of $h(p)$ for all 144,449,537 primes below $3 \cdot 10^9$ finds no prime with $h(p) \geq 3$ outside the hard classes.

Proof. For $p = 2$ and $p \equiv 3 \pmod{4}$, $h(p) = 1$ by Lemma 4.1. Let $p \equiv 1 \pmod{4}$.

- If $p \equiv 5 \pmod{12}$, the shell $(p + 3)/4$ is occupied (Section 3, after Proposition 3.3).
- If $p \equiv 13 \pmod{24}$, the same shell is occupied (Proposition 3.4, first line of the proof).
- If $p \equiv 1 \pmod{24}$ is not hard, Proposition 3.4 shows that the shell with $R = 3$ or the shell with $R = 7$ is occupied.

In each case $R \leq 7$, so $h(p) \leq 2$. A strict record after 73, where $h = 2$, has $h \geq 3$. $\square$

Proposition 4.3 (the record primes). *The strict records $p \leq 8,803,369$, with $h(p)$ and the residual R of the least occupied shell, are exactly*

$$(2, 1, 2), (73, 2, 7), (1129, 3, 11), (1201, 6, 23), (21169, 8, 31), (118801, 15, 59), (8803369, 27, 107).$$

For the odd records $R = 4h(p) - 1$; for $p = 2$ the least denominator is 1, so $R = 4 \cdot 1 - 2 = 2$. Moreover $h(806521) = h(118801) = 15$, so 806,521 is not a strict record. There is no strict record between 8,803,369 and $3 \cdot 10^9$. Below $3 \cdot 10^9$, no prime other than 8,803,369 has $h \geq 22$. The primes with $h = 19, 20$ or 21 are

- $h = 21$: 287,567,281, 651,412,441, 2,188,875,529;
- $h = 20$: 778,103,881, 1,749,233,641, 2,535,613,921, 2,809,257,889;

- $h = 19$: 230,089,729, 793,592,809.

Status. Reproduced here: every prime up to 8,803,369 was computed by Lemma 4.1 and Theorem 3.1, using the residues of the divisors of a^2 modulo R (script `esr_records.py`, under three seconds).

- *Source of the list.* Ionascu and Wilson print the record list [3]. The archive (Proposition 17.6, pp. 131–132) recovers every printed record prime from the definition. It corrects one printed class label: $1201 \in C_6 \setminus C_5$, where the source prints C_4 . Its table (IW-RC) (p. 133) gives $R = 2$ for $p = 2$.
- *The range to $3 \cdot 10^9$.* It was found by the referee pass of this reader with its own program, and recomputed here by an independent segmented program. The two agree on every record, on every prime with $h \geq 19$, and on the histogram of h .
- *The shell at p_* .* At $p_* = 8,803,369$ the least occupied shell has $R = 107$, the shell of archive §3.

For the five records whose residual R is a safe prime ($R = 2m + 1$ with m prime), and whose shell $a = (p + R)/4$ has exactly one non-residue prime factor modulo R , to exponent one, the archive defines a deficit set $D \subset \mathbb{Z}/m$ from the discrete logarithms of the prime factors of a (§19.1). It states the *strict-record deficit-box conjecture* (§19.2): at such a record with non-empty deficit, D is a difference-proper generalised arithmetic progression of rank at most two, and two induced “odd sheets” cover $\mathbb{Z}/m$. The deficits of the first three rows are empty. The two non-empty ones are proved to be a rank-two rectangle $\{11, 14, 15, 18\} \subset \mathbb{Z}/29$ at (118801, 59) and the progression $\{23 + 42j : 0 \leq j \leq 9\} \subset \mathbb{Z}/53$ at (8803369, 107). The boundary example $p = 806,521$ satisfies every hypothesis except strict recordness, and both conclusions fail there. So the hypothesis cannot be dropped.

Status. The conjecture is open, as the archive says. Its finite evidence is reproduced here from the definitions (script `esr_deficits.py`, with the primitive root $g = 2$ that the archive uses for $R = 59$ and $R = 107$):

- the deficits of the rows $p = 73$, 1129 and 1201 are empty, and the row $p = 21169$ is outside the domain because $m = 15$ is composite;
- at $p = 118,801$: $a = 29715 = 3 \cdot 5 \cdot 7 \cdot 283$, $D = \{11, 14, 15, 18\}$, $D - D = \{3r + 4s : |r|, |s| \leq 1\}$ with nine elements, offsets $u = 11$ and $v = 17$, and the two odd sheets cover $\mathbb{Z}/29$;
- at $p = 8,803,369$: $a = 2,200,869 = 3^2 \cdot 11^2 \cdot 43 \cdot 47$, $D = \{23 + 42j : 0 \leq j \leq 9\}$, $D - D = \{42k : |k| \leq 9\}$ with nineteen elements, offsets 29 and 23, and the sheets cover $\mathbb{Z}/53$; also the corollary of archive §16, that the middle target -1 is reached by exactly two exponent vectors and the exterior target not at all;
- at $p = 806,521$: $a = 201,645 = 3^2 \cdot 5 \cdot 4481$, a deficit of 14 points with $D - D = \mathbb{Z}/29$, and the odd sheets miss exactly $\{1, 4, 8, 12, 16, 20, 24, 27\}$.

Another primitive root multiplies D by a unit of $\mathbb{Z}/m$ (archive §19.1). The script confirms that the sizes of D and $D - D$, and the number of missed points, do not change. The failure of conclusion (i) at 806,521 follows from $D - D = \mathbb{Z}/29$ by the archive’s counting argument. A difference-proper progression of rank one with 14 terms has exactly 27 differences. A proper one of rank two has side lengths $\{2, 7\}$, so a difference-proper one would have $3 \cdot 13 = 39 > 29$ distinct differences. That argument was read and is correct. By Theorem 3.1(b), the counts at $p_* = 8,803,369$ ($|E_a| = 0$, $|M_a| = 2$) mean that exactly one unordered solution has least denominator 2,200,869, namely the archive’s (405) (p. 97), from the middle divisor $u = 11^2$:

$$\frac{4}{8803369} = \frac{1}{2200869} + \frac{1}{181085300330} + \frac{1}{3293760527702370}.$$

The archive’s Lean module covers the $R = 107$ row only, as the archive states.

4.2 Fixed-divisor families, and four progressions on which the equation is solved

Theorem 3.1(b) turns a single divisor into a solution. If the divisor is a fixed multiple of the cofactor $h = a/c$, the solution is given by polynomials in n on a whole residue class. The archive states this for primes, as its constant-factor carriers (Lemma 24.218, p. 422). The form below holds for all n . It was found by the referee pass of this reader and checked here.

Proposition 4.4 (fixed-divisor families). *Let $R \geq 3$ be odd, and let $c, s \geq 1$ with $\gcd(c, R) = 1$ and $s \mid c^2$. Let $n \geq 1$ with $n \equiv -R \pmod{4c}$, and put $a = (n + R)/4$ and $h = a/c$, both integers.*

(i) Exterior. If $n \equiv -cs^{-1} \pmod{R}$, then $4/n = 1/a + 1/y + 1/z$ with the positive integers

$$y = \frac{n(nsh + a)}{R}, \quad z = \frac{c^2h/s + na}{R}.$$

(ii) Middle. If $n \equiv -4s \pmod{R}$, then $4/n = 1/a + 1/y + 1/z$ with the positive integers

$$y = \frac{n(a + s)}{R}, \quad z = \frac{n(a + a^2/s)}{R}.$$

Each case is one residue class modulo $4cR$. If $\gcd(cR, 35) = 1$, that class meets all six hard classes modulo 840 or none of them. It meets all six exactly when it is $\equiv 1 \pmod{\gcd(4cR, 24)}$.

Proof. Put $S = na$. In both cases n is a unit modulo R , because $\gcd(cs, R) = 1$. So is a , because $4a \equiv n$. Hence S is a unit modulo R . The proof of Theorem 3.1(b) uses only this. A positive divisor d of S^2 with $d \equiv -S \pmod{R}$ therefore gives positive integers $y = (d + S)/R$ and $z = (S^2/d + S)/R$ with $1/y + 1/z = R/S$. Then $1/a + 1/y + 1/z = (n + R)/(na) = 4/n$.

- In case (i), take $d = n^2u$ with $u = sh$. Here $u \mid c^2h^2 = a^2$, and $d \equiv -S$ means $nu \equiv -a$, that is, $4u \equiv -1 \pmod{R}$. That holds because $c(4sh + 1) = s(n + R) + c \equiv sn + c \equiv 0 \pmod{R}$. Then $S^2/d = a^2/u = c^2h/s$.
- In case (ii), take $d = nu$ with $u = s$. Here $s \mid c^2 \mid a^2$, and $d \equiv -S$ means $s \equiv -a$, that is, $4s \equiv -n \pmod{R}$. Then $S^2/d = na^2/s$.

Since R is odd and prime to c , the congruences modulo $4c$ and modulo R define one class modulo $4cR$ (Chinese remainder theorem). If $\gcd(cR, 35) = 1$, then $\gcd(4cR, 840) = \gcd(4cR, 24)$. A class r modulo $4cR$ meets a class t modulo 840 exactly when $r \equiv t \pmod{\gcd(4cR, 840)}$. The six hard classes are all $\equiv 1 \pmod{24}$. $\square$

Status. Proved here. For primes it is archive Lemma 24.218; the archive's proof of its Theorem 24.219 identifies the four families below as these middle and exterior channel maps.

Theorem 4.5 (four unconditional families). For $k \geq 0$ put $g_1 = 31k + 30$, $g_2 = 31k + 15$, $m_3 = 23k + 15$, $m_4 = 23k + 19$, $w_3 = 1116k + 727$ and $w_4 = 4464k + 3683$. Then $4/n_i = 1/a_i + 1/b_i + 1/c_i$ with positive integers

i	n_i	a_i	b_i	c_i
1	$248k + 209$	$2g_1$	$2n_1(k + 1)$	$2n_1g_1(k + 1)$
2	$248k + 89$	$2g_2$	$n_2(2k + 1)$	$2n_2g_2(2k + 1)$
3	$17112k + 11137$	$186m_3$	$12n_3m_3w_3$	$124m_3w_3$
4	$17112k + 14113$	$186m_4$	$6n_4m_4w_4$	$31m_4w_4$

No primality of n_i is needed.

Status. Proved here by direct expansion: $4a_ib_ic_i - n_i(b_ic_i + a_ic_i + a_ib_i)$ is the zero polynomial in k for each i (sympy), and the entries are positive for $k \geq 0$; exact fractions agree for $k \leq 200$ (script `esr_families.py`). Archive Theorem 24.219, p. 423.

- As *fixed-divisor families*. The four families are instances of Proposition 4.4. With $(R, c, s) = (31, 2, 2)$ and $(31, 1, 1)$ in case (ii), and $(23, 186, 18)$ and $(23, 186, 36)$ in case (i), where $h = m_3$ and $h = m_4$, its formulas give exactly the table (checked by expansion). Two of the families are halves of larger instances (archive Proposition 24.222(ii)–(iii), p. 425):
 - family 2 is half of the class $n \equiv 89 \pmod{124}$, which is $(31, 1, 1)$ itself;
 - family 3 is half of the class $n \equiv 2581 \pmod{8556}$, which is $(23, 93, 9)$ in case (i).

- *Rosati*. The archive shows that these triples are specialisations of Rosati’s two 1954 parametrisations. It claims no independence from the classical parametrisations (archive abstract, p. 2, and §13.1). That comparison is located, not checked.

Proof. All displayed linear factors have positive constant terms, so every entry is a positive integer for $k \geq 0$. The identity $4a_i b_i c_i = n_i(b_i c_i + a_i c_i + a_i b_i)$ is a polynomial identity in k , verified by expanding both sides. Dividing by $n_i a_i b_i c_i$ gives the decomposition. $\square$

Each progression meets all six hard classes modulo 840. This is the last part of Proposition 4.4, and directly: since $\gcd(248, 840) = 8$ and $\gcd(17112, 840) = 24$, the numbers n_1, n_2 run through every class $\equiv 1 \pmod{8}$ modulo 840, and n_3, n_4 through every class $\equiv 1 \pmod{24}$. Every hard class is $\equiv 1 \pmod{24}$, being the square of a unit. Each $n_i(0)$ is prime to the step of its progression, so by Dirichlet’s theorem every progression contains infinitely many primes in each hard class. Examples are the strict records $1201 = n_1(4) \equiv 361$ and $21169 = n_2(85) \equiv 169 \pmod{840}$. So the families solve the equation at infinitely many primes in every class that the reduction modulo 840 leaves open (script `esr_families.py` lists the least prime for each family and class). This does not conflict with the obstruction to polynomial identities recalled in Section 1. That obstruction is modulo q , and it applies only when r is a square modulo q : here $11137 \equiv 5 \pmod{23}$, and 5 is not a square modulo 23. The same holds for the other rows, since $14113 \equiv 14 \pmod{23}$, $209 \equiv 23 \pmod{31}$ and $89 \equiv 27 \pmod{31}$ are non-squares. Proposition 4.6 below shows that this is forced: no fixed-divisor family covers a square class.

4.3 The carrier census

The largest part of the archive’s arithmetic is a sequence of *sufficient-carrier censuses*. It occupies §18 and §24 from the base of the census (Theorem 24.100, p. 335) to the end of §24.3 (p. 477).

- *Domain*. The census concerns only the primes $p \equiv 25 \pmod{72}$ whose shells $R = 7$ and $R = 11$ are both empty (archive Corollary 18.17 and Theorem 24.100). Such a prime lies in one of 30 classes modulo $11088 = 16 \cdot 9 \cdot 7 \cdot 11$, the base rows $B_1 \cup B_{25}$. The primes $p \equiv 1$ and $p \equiv 49 \pmod{72}$ with these two shells empty are outside the census. Below $3 \cdot 10^6$ there are 2,316 primes $p \equiv 1 \pmod{24}$ with both shells empty; 829 are $\equiv 1$, 682 are $\equiv 25$ and 805 are $\equiv 49 \pmod{72}$ (computed here).
- *Rows*. The archive considers residue data of p modulo a growing product of moduli. We call each such datum a *row*. The moduli are 11088, 5, 19, 23, 31 and 47, then 71, 191, 83, 167 and 239.
- *Forced rows*. A row is *forced* when an explicit construction shows that, for every prime p in the branch with that row, one of a finite list of shells is occupied. An example is the implication (2272) on p. 431: for a prime $p \equiv 25 \pmod{72}$ whose shells $R = 7$ and $R = 11$ are empty and whose row lies in the domain but outside the survivor set, some shell with $R \in \{15, 19, 23, 31, 47\}$ is occupied.
- *Survivors*. The remaining rows are *survivors*. Nothing is concluded for them.

The archive’s front section (pp. 3–5) reports the following chain.

Step	Domain (rows)	Forced	Survivors
six shells after the unit lift mod 71	4,590,432,000	4,212,008,640	378,423,360
$R = 191$	71,900,438,400	24,762,018,720	47,138,419,680
$Q = 83$	3,865,350,413,760	934,648,428,742	2,930,701,985,018
$R = 167$	486,496,529,512,988	115,851,891,806,170	370,644,637,706,818
$R = 239$	88,213,423,774,222,684	20,524,419,041,579,843	67,689,004,732,642,841
refresh of the $R = 167$ coefficient		39,025,685,947,842	67,649,979,046,694,999

Cumulatively 2,757,919,929,517,785,001 of the 2,825,569,908,564,480,000 rows of the final hard domain are forced, a proportion of 0.97606 The table is internally consistent (checked here):

- each domain is the previous survivor count times $q-1$, for the next modulus $q = 191, 83, 167, 239$;
- the final domain is $4,590,432,000 \cdot 190 \cdot 82 \cdot 166 \cdot 238$;
- forced plus survivors equals the domain in every row;
- the fractions 69643/75900 and 6257/75900 of the first row are exact.

Status. Located. The arithmetic consistency of the reported numbers was checked here; the carrier theorems and the enumerations behind them were not re-run. The archive states the following about its own checks.

- The enumerations are replayed by independent Python programs.
- Bounded Lean modules check the literal census rows, totals, fractions, target residues and candidate orders, but not the affine and gcd reconstructions, carrier membership or the positive-denominator witnesses (pp. 3–5).
- The ranking that selected 239 after 167 was corrected: on the full ten-coordinate object, 59 ranks first and 239 second. The $R = 239$ step is kept as a valid non-greedy continuation, not as an optimal one (p. 4).

The archive calls these “bounded sufficient-carrier censuses” and states that they are not shell-exhaustion, density, endpoint or optimality theorems (p. 2). A survivor row is only outside the carriers’ images; it is not shown to be shell-empty.

What the census can and cannot do. Every carrier of the census is a fixed-divisor family in the sense of Proposition 4.4. On its row it fixes the residual R , a divisor c of a that the row determines, and a divisor s of c^2 , and uses $u = sh$ or $u = s$:

- for $R = 15, 19, 23, 31$ and for the factor-31 carrier at $R = 23$, the maps listed in the proof of archive Theorem 24.228 (pp. 432–433), with $c = 2, 1, 6$ or $12, 2$ and 186 ;
- for $R = 47$ and later, the sets $E_q(c) = \{-cs^{-1}\}$ and $M_q(c) = \{-4s\}$ over the divisors $s \mid c^2$, with $c = \gcd(a, N)$ read off the row (archive (2268)–(2269), p. 431, and the coefficients c_Q , pp. 462–463).

In each case $4cR$ divides the modulus of the row. The obstruction of Schinzel and Yamamoto therefore applies, in the following form.

Proposition 4.6 (no fixed-divisor family covers a square class). *In the setting of Proposition 4.4, the class modulo $4cR$ on which (i) or (ii) applies contains no square modulo $4cR$. So it contains no class that is a square modulo a multiple of $4cR$.*

Proof. Suppose n lies in the class and is a square modulo $4cR$.

- *The prime 2.* n is odd, so $n \equiv 1 \pmod{4}$ and $R \equiv -n \equiv 3 \pmod{4}$. If $2 \mid c$, then $8 \mid 4c$, so $n \equiv 1 \pmod{8}$, $R \equiv 7 \pmod{8}$ and $(2/R) = 1$.
- *Odd primes $\ell \mid c$.* $-R \equiv n$ is a non-zero square modulo ℓ , so $(-R/\ell) = 1$. By reciprocity, $(\ell/R) = (R/\ell)(-1)^{(\ell-1)/2} = (-R/\ell) = 1$, since $(R-1)/2$ is odd.

Hence $(c/R) = 1$, and $(s/R) = 1$ because $s \mid c^2$. But n is a unit square modulo R , so $(n/R) = 1$. In case (ii), $n \equiv -4s$ gives $(n/R) = (-1/R)(s/R) = -1$. In case (i), $n \equiv -cs^{-1}$ gives $(n/R) = (-1/R)(c/R)(s/R) = -1$. Either way this is a contradiction. $\square$

Status. Proved here; found by the referee pass of this reader. It agrees with the archive’s own residue sets (checked here):

- at $R = 47$ the forced residues F_{36} are exactly the 23 non-residues modulo 47, and F_{18} consists of 21 of them;

- the unforced sets contain every quadratic residue of their modulus: $S_5 = \{1, 4\}$, $A_{25} = Q_{23}$, and S_{19} and $S_{31}^\#$, whose excluded residues 15, 18 mod 19 and 15, 23, 27, 29, 30 mod 31 are all non-residues.

Corollary 4.7 (the census cannot finish). *No row that is a square modulo the modulus of the census is ever forced by its carriers. All 30 base rows are squares modulo 11088. So the rows whose coordinates at the ten odd primes 5, 19, 23, 31, 47, 71, 191, 83, 167, 239 are all non-zero squares are never forced. They number*

$$\frac{2,825,569,908,564,480,000}{2^{10}} = 2,759,345,613,832,500,$$

that is, 1/1024 of the final domain and 4.08% of its 67,649,979,046,694,999 survivors. Each further odd prime modulus added to the census halves this share, but no finite chain of such carriers removes it.

Proof. A forced row lies in a class of Proposition 4.4 modulo $4cR$, and $4cR$ divides the modulus of the row. Proposition 4.6 applies. The base rows were checked directly: each is a square of a unit modulo 11088 (script `esr_referee_checks.py`). The domain is $30 \cdot \prod (q - 1)$ over the ten odd primes, and exactly half of the units modulo each odd prime q are squares. $\square$

The survivors make up 2.39% of the final domain. Each of the four lifts forced between 23% and 35% of its domain ($R = 191$: 34.4%; $Q = 83$: 24.2%; $R = 167$: 23.8%; $R = 239$: 23.3%). Corollary 4.7 shows that a residue census built from such carriers cannot reach every hard prime. A proof along these lines needs carriers that use more than the residues of p , such as the prime factors of $(p + R)/4$ (compare Proposition 3.6). The archive itself names the removal of the survivors as the open global obligation (p. 4).

5 The September continuation

The continuation of 17–23 September is organised as the 28-item “Start reading” list of the README, with proof packages in `research/incoming/`. The workbench describes the results of its 22 September release as written proofs with Python replays, without a Lean build or independent human review (43a_§2, citing `DAILY_RESULTS_20260922`).

5.1 The audited items

The 28 items of the README’s “Start reading” list (17–23 September) are listed in 43a_§2, with their packages. Those audited:

- *Item 6* (audited). With $L_a = 8\tau(a)^2/\varphi(R_a) - \mathcal{E}_a$, where $\mathcal{E}_a$ is the residue-collision energy of the $2\tau(a)$ divisors of pa modulo R_a : for $p \equiv 1 \pmod{4}$, $p \geq 2^{20}$, $\sum_{a \in (p/4, p/2)} L_a \leq -\frac{1}{24}p \log p$. So the lower bound that keeps only the principal character, summed over the shells, is negative for large p : that method cannot prove occupancy.
- *Item 7* (audited). For $p \equiv 1 \pmod{8}$, bounds in terms of the least quadratic nonresidue; at the hard primes a middle state (a, u) on a shell $p/4 < a < p/2$ has $a \leq 3(p + 3)/11$, and an exterior state satisfies $22(p + 1) \leq 23(p - 2a + 1)^2$. The workbench says this localises solutions and does not prove the region occupied.
- *Item 14* (audited). For $h' = 4j - 1 > t$ the divisors $W \mid j^2$ with $W \equiv t \pmod{h'}$ are canonical, companion or finite templates; the finite ones satisfy the sharp bound $h' \leq t^2 - 3t + 1$, with equality exactly in one parametrised family, attained at infinitely many primes $p \equiv 1 \pmod{840}$. For every $t \geq 4$ infinitely many hard primes have an exterior state whose same-grade middle box is empty.
- *Item 21* (audited). For a proper rational trace source the complete set of same-word residual returns is $\{k \mid R : d \mid k, k \equiv 1 \pmod{4K(u)}\}$; a uniform return exists exactly for the nine words $u \mid 36$.

- *Item 22* (audited, certificate). For every solution at $p \equiv 1 \pmod{12}$ an explicit symmetric polynomial $\mathfrak{N}(p, x, y, z)$ of degree 8 satisfies $\mathfrak{N} < -\frac{125873811}{262144}p^8$ ($\approx -480.17p^8$), and $\mathfrak{N} < -6960.19p^8$ when $p \equiv 1 \pmod{24}$; the proof reduces to finite polynomial positivity certificates, recomputed independently. It is a statement about existing solutions, not about their existence.
- *Item 26* (audited in part: the least example and the deduction from it; its prime-square trace theorem at $p \equiv 1 \pmod{24}$ and its terminality statement read and correct as read, with the identity $16(4q^3+1)-(4q+1)(16q^2-4q+1) = 15$ checked; the factor-sum, Pell and fixed-tail-sum sections not checked). $p = 67369$ is the least prime $p \equiv 1 \pmod{24}$ whose first “trace shell” ($a = 16849, R = 27$) precedes its first full shell ($a = 16850, R = 31$); so no universal map from rational traces to solutions avoids increasing a .
- *Item 4* (conditional on a computer enumeration of 1,381,117,764 rooted profiles, 5,922,259 of them range-compatible, run by two implementations of the workbench and not re-run here). At every hard prime, from any nonresidue prime, factoring at most 25 numbers $(p + \sigma q t^2)/4$ with $t \in \{1, 3, 5, 7, 9\}$ yields at least six distinct nonresidue primes. A direct test of all 108 hard primes below 40,000 agrees.

Items 2 and 3 are method counterexamples, and their certificates are audited: the seven-cycle sink of item 2 at $p = 2521$ and the closed triangle of item 3; their closure and escape theorems are located. For item 8, the middle identity $p = RQ - (Q + 1)^2/(4A)$ and its two bounds were reproduced on all 1,613 oriented middle states at primes $p \equiv 1 \pmod{8}$ below 4000, and its atlas is located. The 22-digit prime 7510085481569082811681 of item 3 is proved prime here by a Pocklington chain ($p - 1 = 2^5 \cdot 3 \cdot 5 \cdot q_1, q_1 - 1 = 2 \cdot 3 \cdot 5 \cdot 351707 \cdot q_2, q_2 - 1 = 2 \cdot 3 \cdot 17^2 \cdot 5051 \cdot 169307$, witnesses below 12 at every level, the leaves by trial division). Item 23 is audited in part (its certificates; the deduction read in part); items 11–13, 15 and 20 use imported material (Section 7); the remaining 17 items, including these five, are located. The results bench R01–R10 and the arguments X1–X7 are checked in 43a_ §3; one source passage error (X5: $289^3 \equiv 169$, not -1 , modulo 840) is recorded by the workbench itself.

5.2 The $(3, 4, \infty)$ monodromy covers

A package of the workbench (ES-09) imports three integer matrices A_1, A_2, A_∞ with $A_1^3 = A_2^4 = A_1 A_2 A_\infty = I$ from the S^6 manuscript [1] and nothing else from it. For odd levels D , which is the source’s standing hypothesis, the induced affine action on $(\mathbb{Z}/D)^3$ is transitive when $3 \nmid D$ and has two orbits, of sizes $D^3/9$ and $8D^3/9$, when $3 \mid D$; for $3 \nmid D$ the cover is connected of genus $(5D^3 - 12D^2 - 17D + 24)/24$, and for $3 \mid D$ the two orbits give components of genera $(5D^3 - 12D^2 - 81D + 216)/216$ and $(5D^3 - 12D^2 - 9D + 27)/27$. These statements are false for even D (at $D = 2$ the orbits have sizes 2 and 6; found by the referee of the notes), which the ES application never uses, since there D divides the odd number R_a . At the levels $D = 2^k$ with $1 \leq k \leq 5$ the action has exactly two orbits, of sizes $D^3/4$ and $3D^3/4$. The genera of the two components are $(0, 0)$, $(1, 3)$, $(17, 53)$, $(177, 537)$ and $(1569, 4721)$. This was found by the referee pass of this reader and recomputed here from cycle counts, with the matrices of the package’s `src/core.tex`. The package also shows that the equation fails at p if and only if a denominator-labelled cover has no section. That equivalence is correct, and it is Theorem 3.1 again: a section exists exactly when the tail denominator $R_a / \gcd(R_a, 4u + 1)$ or $R_a / \gcd(R_a, u + a)$ equals 1.

Status. Audited: the proof read; the relations, the orbits and the genera also recomputed from cycle counts for $D = 1, 3, \dots, 15$. The equivalence is a re-encoding of Theorem 3.1; the assessment is this audit’s (the package presents it as a complete failure object).

6 Negative results, with exact scope

Each row says which proposed implication fails and what the failure does not show. Every certificate prime in the table is itself solvable.

Source	What fails	Certificate	What it does not show
bounded transport	every occupied shell has a solution with unary layer $A \leq 2$	$p = 37, a = 12$: only $u = 8$	the global cutoff ($a = 10$ works at 37)
first-two-shell sieve	occupancy of the $R = 7$ shell gives a first-two-layer hit	$p = 373, a = 95$	anything about other shells
item 2	every canonical closed component contains a local hit	$p = 2521$: a 7-cycle sink	2521 is solvable
item 3	strict canonical descent ends at a hit	a closed triangle at a 22-digit prime	that prime is solvable
item 6	the principal-only bound, summed over shells, is positive for large p	$\sum L_a \leq -\frac{1}{24}p \log p$	anything about the true counts
item 6	a nonnegative reweighting of it certifies occupancy	$p = 87481$: all 21,870 shells have $L_a \leq 0$	only this certificate family fails
item 14	same-grade middle repair from exterior states with $\alpha = -4t, t \geq 4$	$p = 7840561$	those primes are solvable
item 21	a same-word residual return for words $u \nmid 36$	3,041 of the 3,654 sources with $u \nmid 36$ at primes below 2600	endpoint solutions exist
item 26	a trace-to-solution map never increases a	$p = 67369$	67369 is solvable
record shells (archive §19.3)	the deficit-box conclusions without strict recordness	$p = 806,521$: full difference set, eight points missed	anything about strict records
Jacobi barrier (archive §10)	its converse: a non-residue prime factor of a makes the shell occupied	$p = 5, R = 7, a = 3$	the barrier itself (Proposition 3.6)
no-two-shears theorem (supplement §5)	the theorem at the primes $p \equiv 2 \pmod{3}$ with $R(R+4) \mid p+4$ for a suitable residual R , for instance at every $p \equiv 131 \pmod{180}$; so no congruence hypothesis weaker than $p \not\equiv 2 \pmod{3}$ suffices (Propositions 2.1 and 2.2)	two-edge paths at $p = 131$ and $p = 1613$	anything at $p \not\equiv 2 \pmod{3}$, where the theorem holds; it holds at most primes $p \equiv 2 \pmod{3}$ (Remark 2.3); anything about solvability
carrier census (archive §24)	forcing every row of the census domain by fixed-divisor carriers	the 2,759,345,613,832,500 square rows (Corollary 4.7)	anything about the primes in those rows; carriers that use the factorisation of $(p+R)/4$

All of these are statements about methods or certificate families. None is a counterexample to the conjecture, and none bears on whether it is true.

The archive records further negative results of the same kind. They are located, not checked here:

- the withdrawn constructions of §5 (p. 22), each with its point of failure;

- the retracted converse of the Star–Kneser forcing theorem (§15.3, p. 122);
- the failed universal bridge of Dyachenko’s bounded identities (§12.2, p. 103);
- a finite counterexample to one projection step in López Bonilla’s 2026 preprint (§19, provenance, p. 148): in C_6 , the set $\{0, 1\}$ has trivial stabiliser, but its reduction modulo 2 is all of C_2 . The archive restricts this to that step and does not assess the whole paper.

7 Where the workbench meets the author’s other workbenches

The owner notes that the Erdős–Straus and zeta sessions often ran at the same time, and that material crossed between them. The note 47_ records every crossing that the audit found as a typed edge. Its edge types are:

- P : a proof-level import;
- D : a dependency or citation;
- I : the same identity or object on both sides;
- A : an analogy only.

The table lists the edges that touch this workbench, and adds three that this reader found in the archive (marked “new”).

Edge	What crosses	Type	Status
NS $\rightarrow$ ES	The auxiliary torus endomorphism $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ (determinant 14) of the Navier–Stokes manuscript. It enters 21 statements of the supplement (§8) as an operator only. The core identity is $\{(u, v) \in G^2 : u^3v = 1 = uv^5\} = \{(u, u^{-3}) : u^{14} = 1\}$	P	audited (21_ §1); no fluid theorem is used
$S^6 \rightarrow$ ES	The integral monodromies A_1, A_2, A_∞ of the $(3, 4, \infty)$ period system, in ES-09	P	audited for odd levels (§5.2)
ES $\leftrightarrow$ S^6	The Niemeier lattice with root system $A_5^4 D_4$ and glue index 72 (ES result R10)	I	ES’s glue code verified; the two codes not compared
Jacobian map	Alpöge’s map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with $\det DF = -2$ and the three-point fibre over $(-\frac{1}{4}, 0, 0)$ (archive §§4, 7). The same map is in the Yang–Mills workbench and the zeta reader. ES’s four-dimensional extension P ($\det DP = -2$) crosses to the zeta programme’s conductor	P	$\det DF$, $\det DP$ and the fibre verified (zeta reader [10], Lemma 6.6; 21_); the conductor maps located
ES $\leftrightarrow$ zeta	The split-zero semiring $G(R) = R \sqcup \{\tau\}$ with its Boolean support character (archive §21; ES R01)	I	audited (21_ §6)
ES $\leftrightarrow$ zeta	A finite Weyl phase-space basis	I	ES side verified on $C_2 \times C_3$
zeta $\rightarrow$ ES	NG20–NG21, used by item 20; the SplitZero packages of 13–16 September	D	located
ES $\rightarrow$ zeta	The 488-page reader’s source, vendored into the zeta repository	D	located
ES $\rightarrow$ zeta	The 124-page ES/Fable reader: solutions as binary quartics, then the Fable cover and a weighted conductor (§2.4)	P	the quartic relation proved here (Proposition 2.4); the rest located
ES $\leftrightarrow$ Collatz	$3(4n + 1) + 1 = 4(3n + 1)$, i.e. $\theta T = 4\theta$ (ES R05, diagrams by u/CivQ17)	I	audited

Edge	What crosses	Type	Status
Collatz $\leftrightarrow$ ES $\leftrightarrow$ zeta	The homogenisation $(x, h) \mapsto (ax + bh, h)$ (ES X4)	I	ES’s diagram read
Busy Beaver (new)	Busy-Beaver extrema quoted by the archive ($S(4) = 107$ and $\text{Space}(5) = 12289$; for the five-state values it cites the formal enumeration of the bbchallenge collaboration) select the shells of archive §§3 and 20; §25 maps extrema to shell carriers. The zeta reader’s detectability argument uses a different machine for a different purpose	A	located; no mathematical dependence between the two
Connes (new)	Supplement §6 reads Connes’ primitive intersections; archive §24 opens with Tomita–Takesaki transport. The zeta programme uses Connes’ summation map and Meyer’s quotient	A	located; shared names, no common statement found
Fable (new)	“Fable” names three different objects across the repositories: the three-dimensional Keller map, the S^6 construction in older files, and a conductor cover (47a_ §10). Archive §5 withdraws one Fable construction	—	disambiguate at each use
ES $\leftrightarrow$ YM	The word “gap” (ES X7), and a proposed success projector	A	no mathematical content

So the mathematical crossings are few and specific: the torus operator, the monodromy matrices, the Keller maps, the split-zero semiring, and two identities. Each is either audited or recorded as located. The remaining connections are shared vocabulary or proposals. The archive says the same of its late Fable, operator, Connes and moonshine material (p. 4): it is admitted only as a candidate interface, pending an explicit typed map and compatibility check.

8 What remains open, and what was not checked

8.1 Open questions

1. *The global obligation.* Show that for every prime $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$ some shell is occupied. This is the conjecture itself, restated by Theorem 3.1. The archive names it as the remaining obligation (p. 4). By Propositions 3.6 and 3.7, any proof must produce, for some R , a shell whose group G_a contains -1 . When R is prime this means a prime factor of $(p + R)/4$ that is a non-residue modulo R . The proof must then go further, since that condition alone does not suffice; Proposition 3.8 says when it does.
2. *The census survivors.* 2.394% of the final hard domain survives the carrier chain (§4.3).
 - The first half of the question asked before the referee pass is now answered. The census’s carriers can never force the 2,759,345,613,832,500 square rows, and no finite chain of such carriers can (Corollary 4.7).
 - Open: can the non-square survivor rows all be forced by carriers of the same kind, if more moduli are added?
 - Open: what kind of carrier, using the factorisation of $(p + R)/4$ and not only residues of p , could force the square rows?
3. *The strict-record deficit-box conjecture* (§4.1; archive §19.2). It holds at the two strict records with non-empty deficit below 8,803,369 (reproduced here), and strict recordness cannot be dropped

from it ($p = 806,521$). A further test needs a strict record above 8,803,369. There is none below $3 \cdot 10^9$ (Proposition 4.3).

4. *Growth of the least class.* Is h unbounded on the primes?

- Along the strict records up to $3 \cdot 10^9$, h takes the values 1, 2, 3, 6, 8, 15, 27.
- Below $3 \cdot 10^9$ the next largest values are 21, 20 and 19, at nine primes (Proposition 4.3).
- Every record after 73 lies in a hard class (Corollary 4.2).
- The workbench states no conjecture on the growth of $h(p)$, and none is stated here.

5. *The density of the shear failures.* What is the exact relative density, among the primes $p \equiv 2 \pmod{3}$, of the primes with a path of two shear edges? It is at least $1/24$, from the class $131 \pmod{180}$, and at most 0.335 (Remark 2.3). Below 10^7 the proportion is 5.24%.

6. *The $(3, 4, \infty)$ covers at even levels.* At $D = 2^k$ with $k \leq 5$ there are two orbits, of sizes $D^3/4$ and $3D^3/4$ (§5.2). Does this hold for every k ? Is the orbit decomposition at a level $D = 2^k m$ with m odd the product of those at 2^k and at m ? At $D = 12$ it is: the orbits have sizes 48, 144, 384 and 1152, the products of 16, 48 and 3, 24.

8.2 What this reader did not check

- *The archive.* Apart from the parts marked proved or reproduced in §2.2, its 619 pages were mapped, not audited. In particular, the side constructions (§§2, 4, 6–8, 9 after its opening, 21–23, the operator part of §24, the side material of §24.4, and §25) and the literature audits (§§11–14 and 17) were not compared with the papers they audit.
- *Lean.* The source archive contains 67 files with the extension `.lean` (seven of them `.axioms.lean`); they were not rebuilt. The archive’s own account of what each Lean module checks is quoted where it matters (§4.3).
- *The census.* The enumerations of §4.3 were not re-run. Only the arithmetic consistency of the reported counts was checked, together with the base rows and the carrier sets used in Corollary 4.7.
- *The supplement.* The general proofs of the supplement’s Sections 1–7 were not re-derived, except that of the no-two-shears theorem of Section 5 (Proposition 2.1); their finite content was reproduced in part (§2.3). Section 8, the auxiliary torus, was not checked.
- *The 124-page reader.* Only Proposition 2.4 was checked.
- *The continuation.* Items 1, 5, 9–13, 15–20, 24, 25, 27 and 28 were located, not checked. So were item 4’s enumeration, the deduction in item 23, and ES-04 to ES-08 beyond the ES-08 witness.
- *Novelty.* No literature search for novelty was made here. The workbench claims no novelty for the classical coordinates and identities it uses. The results marked as found by the referee pass (Propositions 2.2, 3.7, 3.8, 4.4 and 4.6, and Corollaries 4.2 and 4.7) are short consequences of classical tools. Some may be known; none is claimed as new mathematics.

A Catalogue of statements

This catalogue lists every statement of the workbench that this reader states, proves, reproduces or locates. For each it gives where the statement is in the workbench and how far it was checked here, in the labels defined in “About this reader”. “WB” quotes the workbench’s own label where it adds something. Locators are as in the front matter: “archive” is the 619-page PDF, “supplement” the 86-page supplement, and “item n ” the n -th entry of the README’s “Start reading” list.

Statement	Where in the workbench	Status here	Here
<i>The shell reduction</i>			
The ordered pairs completing a first denominator a number $2 E_a + M_a $; solvable iff some shell is occupied (ES-01, ES-02)	archive Theorems 9.1 (p. 38) and 9.64 (p. 90), table (394); supplement §1 (for $p = 12h + 1$)	Proved here; the bijection and the count are the archive's, for every odd prime; this reader adds the swap analysis and (c)	Thm 3.1
The least denominator lies in $(p/4, p/2)$, except for one solution at $p \equiv 3 \pmod{4}$	a reduction step of item 22	Proved here, for every odd prime	Thm 3.1(a)
A counterexample prime has $E_a = M_a = \emptyset$ at every shell (the all-shell no-hit sieve)	supplement §1	Audited (43a_§1.2); it is Thm 3.1(c)	§3
The residual-3 shell: criterion and count	supplement §1; archive Theorem 10.3	Proved here; count extended to $p \equiv 5 \pmod{12}$	Prop. 3.2
The residual-7 shell: criterion	supplement §1; archive §10	Proved here, for every $p \equiv 1 \pmod{4}$	Prop. 3.3
Both first shells empty only if p lies in a hard class modulo 840	archive Corollary 10.7 (p. 95); the first pass observed the congruence modulo 24	Proved here	Prop. 3.4
Reduction to the six hard classes modulo 840	archive Theorem 10.6, p. 95 (with shells $R \leq 7$); classical (five identities)	Proved here	Props. 3.4, 3.5
Jacobi-symbol barrier, and the failure of its converse	archive Theorem 10.9 and Proposition 10.10, pp. 96–97	Proved here	Prop. 3.6
The group barrier $-1 \notin \langle 4, \ell : \ell \mid a \rangle$, and a converse under an exponent condition	implied by archive §10	Proved here (found by the referee pass)	Props. 3.7, 3.8
<i>Least classes, records and deficits</i>			
$h(p) = (R + 1)/4$ for the least occupied shell	archive §§17.4, 19	Proved here	Lemma 4.1
$h(p) \leq 2$ outside the six hard classes; every strict record after 73 is hard	implied by archive Theorem 10.6	Proved here (found by the referee pass); checked to $3 \cdot 10^9$	Cor. 4.2
The seven strict records up to 8,803,369; $1201 \in C_6 \setminus C_5$; $h(806521) = 15$	archive Proposition 17.6, pp. 131–132 (recovering Ionascu–Wilson's printed list and correcting one label); §19.2, p. 152	Reproduced	Prop. 4.3
No strict record in $(8,803,369, 3 \cdot 10^9]$; the primes with $h = 19, 20, 21$ there	—	Reproduced (two independent programs)	Prop. 4.3
At $p_* = 8,803,369$ the least occupied shell has $R = 107$, and it carries one unordered solution	archive §3	Reproduced	§4.1
The mod-107 deficit progression, the covering, and the two middle fibres (WB: credited to u/CommonCareful3149)	archive §16; bench R06	Reproduced	§4.1

Statement	Where in the workbench	Status here	Here
The deficits at 118,801 and 8,803,369; the boundary example 806,521	archive §19.3	Reproduced	§4.1
The strict-record deficit-box conjecture	archive §19.2 (WB: conjecture)	Open	§4.1, §8
The shell $a = 3075$ at $p = 12289$: counts (3, 6, 3)	archive §20	Counts reproduced (esr_referee_checks.py); the orbit and the quotient certificate located	§2.2
<i>Families and the census</i>			
Fixed-divisor families: a divisor $u = sh$ or $u = s$ with $s \mid c^2$ solves the equation on a class modulo $4cR$	archive Lemma 24.218, p. 422 (for primes)	Proved here, for all n (found by the referee pass)	Prop. 4.4
Four families solving the equation on $248k + 209, 248k + 89, 17112k + 11137, 17112k + 14113$	archive Theorem 24.219, p. 423; Proposition 24.222	Proved here; instances of Prop. 4.4; that they meet every hard class is shown here	Thm 4.5
The carrier census: 97.606% of the final hard domain forced	archive pp. 3–5 and §24 (WB: bounded sufficient-carrier census)	Located; the arithmetic of the counts checked	§4.3
No fixed-divisor carrier forces a square class; the census cannot force its 2,759,345,613,832,500 square rows	—	Proved here (found by the referee pass)	Prop. 4.6, Cor. 4.7
The residual-eleven support ideal; the paired $R = 7/R = 11$ affine theorem	archive §18	Located	§2.2
<i>The supplement and the 124-page reader</i>			
The finite content of Sections 1–7 (gluing, ratio transport, shears, successors, orchard partition, boundary law, unary box, incidence, raw scale)	supplement §§1–7	Reproduced in part	§2.3
No two shear edges compose	supplement §5 (stated for $p \equiv 1 \pmod{12}$)	Generalised here to $p \not\equiv 2 \pmod{3}$; no weaker congruence hypothesis suffices; fails only on a minority of the $p \equiv 2 \pmod{3}$	Props. 2.1, 2.2, Rem. 2.3
The Navier–Stokes auxiliary cover	supplement §8	Not checked	§2.3
The quartic of a solution: $p = -5u_4/u_3$ and the coefficient relation, and its converse	124-page reader; item 11	Proved here (also 21_); the converse found by the referee pass	Prop. 2.4
The Fable cover and the weighted conductor	124-page reader	Located	§2.4
<i>The September continuation</i>			
Item 2: a seven-cycle sink at $p = 2521$ (WB: method counterexample)	research/incoming/	The sink audited (certificate); the closure theorems located	§5, §6
Item 3: a closed triangle at a 22-digit prime	research/incoming/	The triangle audited (certificate), its primality proved here; the escape theorem located	§5

Statement	Where in the workbench	Status here	Here
Item 4: six nonresidue primes within 25 factorisations (WB: computer-assisted)	research/incoming/	Reductions read; conditional on an enumeration not re-run	§5
Item 6: the principal-only bound is negative in sum	research/incoming/	Audited	§5
Item 7: least-nonresidue localisation	research/incoming/	Audited	§5
Item 8: the middle identity and its bounds	research/incoming/	Reproduced on 1,613 middle states; the atlas located	§5
Items 14 and 21: template bound; same-word returns	research/incoming/	Audited	§5
Item 22: the degree-8 invariant is negative on solutions	research/incoming/	Audited (certificate)	§5
Items 23 and 26	research/incoming/	Audited in part	§5
Items 1, 5, 9–13, 15–20, 24, 25, 27, 28	research/incoming/	Located (item 11’s quartic proved here)	43a_ §2
ES-09: the $(3, 4, \infty)$ monodromy covers	received/es_s6_counterfactual/	Audited for odd levels; false for even levels; the levels 2^k , $k \leq 5$, computed	§5.2
ES-08: a forced solution at $p = 1201$, $a = 310$, $R = 39$	12 September continuation	The witness checked; the forcing inequality not audited	43a_ §4
Results bench R01–R10, arguments X1–X7	RESULTS.md	Checked in 43a_ §3 (X1 not recomputed)	§5
<i>The rest of the archive</i>			
Side constructions: §§2, 4–8, 21–23, 25 and the operator part of §24	archive	Located; §5 read; the fibre of §7 verified in the zeta reader	§2.2, §7
Literature audits: §§11–14 and 17	archive	Located; the classes of §17 replayed	§2.2
The Star–Kneser forcing theorem and its retracted converse	archive §15	Located	§6

B How this reader was checked

Sources.

- The repository KokunoYumeto/erdos-strauss-foundation at commit d20b32e, read-only.
- The 619-page archive and its source archive, downloaded on 26 September 2026 from the Zenodo record 10.5281/zenodo.22678971:
00_ERDOS_STRAUSS_Project_Reader.pdf MD5 c1ee553967e0e6f12d4976e171c11b15
01_ERDOS_STRAUSS_Source_Verification_and_Machine_Memory.zip MD5 4129ec50d0c02ff7daf61fde6c589de4
The main source file of the archive (60,561 lines) and its fourteen satellite files were read where they are cited.
- The supplement’s source, in the repository folder research/expanded-2026-09-08/exact_bounded_transport_72/.
- For the literature statements of §1: the source of Elsholtz and Tao’s paper (arXiv 1107.1010v6), which the repository holds in its literature folders.

What was refereed earlier. Theorem 3.1 and Propositions 3.2–3.5, the audited items of Section 5, and Section 5.2 come from the Erdős–Straus section of the reader of four workbenches. There they went through five stages:

1. a first-pass audit by a separate Claude instance, which read the workbench and wrote independent checks (43a_);
2. a referee pass on the notes 43_–47_, by an instance that had written none of them; it found, among other things, that the orbit statements of Section 5.2 fail at even levels;
3. a referee pass on that reader;
4. two further passes that read it for understatement and missed generality as well as for overstatement.

These passes produced the range $x < p/2$ in Theorem 3.1, the first form of Proposition 3.4, Proposition 3.5, and the extension of the count of Proposition 3.2 to $p \equiv 5 \pmod{12}$. The checks are `checks/workbenches/workbench_reader_claims_checks.py` and `checks/workbenches/generality_checks.py`; the referee reports are in `checks/workbenches/`.

What this reader adds to the reader of four workbenches. “New” here means new relative to that earlier reader, not new mathematics. Where the archive already has a statement, the reader cites it.

- Proposition 3.6, which is the archive’s Theorem 10.9 and Proposition 10.10, re-proved;
- Lemma 4.1 and Proposition 4.3, with the reproduction of the deficit data in §4.1;
- Theorem 4.5, and the statement that each family meets every hard class;
- Proposition 2.1;
- Proposition 2.4(a), whose relation was also checked in 21_;
- the map of the archive (§2.2), the consistency of the census counts (§4.3), and the three new rows of the overlap table (Section 7).

These went through the referee pass described next.

The referee pass of this reader. An independent Claude instance (claude-opus-5-5), which wrote none of the text, refereed version 1 of this reader on 27 September 2026. It read in both directions, as the owner asked: for statements stronger than their proofs or sources, and for understatement, missed generality and missed or implied results. It reran the reader’s five scripts, whose outputs were identical, and wrote its own checks. It found every proof correct as written and every recomputed number in agreement. It reported five major and ten minor findings, and seven results that the reader’s material implies. Every finding was verified here before it was applied: by reading the archive at the cited page, or by an independent script.

- *Attributions.* Theorem 3.1(b) is not a generalisation of the archive. The archive’s Theorems 9.1 and 9.64 already hold for every odd prime; the reader now says so. The check here also found that the count $2|E_a| + |M_a|$ is in the archive’s Theorem 9.64, which the referee had credited to the reader.
- *Understatements.* Proposition 3.4 now concludes that the survivors lie in the hard classes modulo 840, as the archive’s Corollary 10.7 does. The census domain is stated. The page map of §24 is split, and the page counts are made consistent.
- *Two misdescriptions.*

- The remark that the census carriers are “not polynomial identities alone” was wrong. They are polynomial identities on their rows, and Proposition 4.6 and Corollary 4.7 replace it.
- Proposition 2.1(b) had suggested that the theorem fails for every $p \equiv 2 \pmod{3}$. It fails only on a minority of them, and Proposition 2.2 and Remark 2.3 now describe that set.
- *Minor findings.* Among them:
 - the residual $R = 2$ at $p = 2$;
 - the archive’s correction of Ionascu and Wilson’s label for 1201;
 - the count of .lean files (67, not 84);
 - the title-page date (30 August);
 - Elsholtz and Tao’s definition of solvability by polynomials;
 - the full citation of Vaughan’s paper.
- *Results adopted.* After verification, these are Corollary 4.2, Propositions 3.7, 3.8, 2.2, 4.4 and 4.6, Corollary 4.7, Proposition 2.4(b), the records to $3 \cdot 10^9$, and the even levels of §5.2.
- *A correction to the referee.* It gave 3,533,141 as the prime with a shear path of length 5. A complete enumeration here shows that the least such prime is 1,183,451.
- *Not adopted.* The referee also reported an observation of Ionascu and Wilson on residues modulo 9240. It could not be checked against their paper here, and is omitted.

The referee’s report and its check programs are in `checks/es_reader/referee/`; the verification scripts of this reader are in `checks/es_reader/referee_verify/`.

The scripts of this reader. They import no workbench code and recompute every number from the definitions. They are in `checks/es_reader/` on the branch named below.

Script	What it checks	Time, result
esr_records.py	$h(p)$ for every prime up to 8,803,369 through Lemma 4.1 and Theorem 3.1; the strict records; $h(806521) = h(118801) = 15$	3 s; as stated
esr_lemma_checks.py	Proposition 3.6 on all 47,210 shells meeting its hypothesis at odd primes below 3000 (none occupied), and the failure of its converse; Lemma 4.1 against a brute-force enumeration of all solutions at the odd primes below 700	25 s; all pass
esr_deficits.py	the deficit data of §4.1 at 73, 1129, 1201, 21169, 118,801, 8,803,369 and 806,521, their independence of the primitive root, the corollary of archive §16, and the counts $ E_a = 0$, $ M_a = 2$ at p_* (22 checks)	1 s; all pass
esr_families.py	the four identities of Theorem 4.5 as polynomial identities, exact fractions for $k \leq 200$, the hard classes met and the least prime in each	1 s; all pass
esr_twoshears.py	Proposition 2.1: the full shear graph of every prime below 2000, and a complete search for two-edge paths up to 10^7	23 s; all pass
referee_verify/ shears_independent.py	Proposition 2.2: degrees on the full graph below 1200; complete enumeration of paths of length 2 to 5 below 10^7 ; the class 131 mod 180	21 s; as stated
referee_verify/ shears_density_ bound.py	the bound $\sum_R \delta_R < 0.335$ of Remark 2.3	5 s; as stated
referee_verify/ hsegment.c, hsegment_A.c	$h(p)$ for all primes below $3 \cdot 10^9$ by a segmented sieve; the records, the primes with $h \geq 19$, and Corollary 4.2	2 min; as stated
esr_referee_checks.py	Propositions 3.4, 3.7, 3.8; the census domain; the base rows and the square-row count of Corollary 4.7; the shell at $p = 12289$; the even levels of §5.2	2 min; all pass

A partial second pass. A second-pass audit of the supplement was started with a separate Claude instance and was stopped by a usage limit before it wrote its report. Its scripts and outputs are kept in `checks/es_reader/second_pass_partial/`, and §2.3 reports what they show. The one reported mismatch was traced to the script’s own expectation (§2.3).

What has not been done. This version has had one independent referee pass. The results adopted from it were proved by the referee and checked here, but they have not had a second independent reading. Nothing in this reader has been reviewed by a human specialist. The parts of the workbench marked located were not checked, and §8 lists what else was left out.

Where everything is. This reader, its sources and its scripts are on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`, in `contrib/claude-ab/grind_20260925/`: the reader in `es_reader/`, the scripts in `checks/es_reader/`, and the audit notes `21_`, `43_`, `43a_`, `47_` and `47a_` at the top level.

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- [10] Claude (Anthropic), *The split-zero programme around the Riemann zeta function: a reader of its results, with proofs*, version 2, Zenodo, 2026, DOI 10.5281/zenodo.22977243 (version 1: DOI 10.5281/zenodo.22970703; Lemma 6.6 has the same number in both).
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